K-Bound: When Is Label-Free Adaptation Knowable?

Helpful, Harmful, and Unknowable Regimes Under Distribution Shift

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Abstract

Test-time adaptation updates a model without new target labels, but an update can help or harm. We ask when the available evidence supports replacing a frozen predictor with a fixed adapted candidate. K-Bound answers this question for a binary population model with a declared bound β on the target calibration residual. We construct target label laws that preserve all label-free observations and derive the exact feasible interval for paired benefit. A strict decision is supported precisely when $|M| > \beta$; inside the closed band, no strict direction is uniformly justified. We also show why the same observations cannot audit away the unresolved residual bound. Knowability-Guided Adaptation (KGA) is an empirical companion, not an implementation of this frontier. It learns to predict observed evaluation-cell benefit from labeled historical outcomes, then uses a residual-based interval to choose ADAPT, FREEZE, or ABSTAIN without new evaluation labels. On an exploratory, cross-fitted CIFAR-10-C panel, Tent and EATA have lower point regret than either fixed policy, while SAR favors always-adapt. On CCT-20, KGA retains frozen predictions and avoids observed degradation from always-adapt, but makes no ADAPT decisions. These results support candidate-specific commitment control; they do not establish interval coverage or population-risk protection on unseen natural shifts.

Keywords: test-time adaptation, safe adaptation, abstention, selective prediction, distribution shift, identifiability, label-free risk estimation

1 Introduction

Test-time adaptation changes a model using unlabeled deployment data. Entropy minimization, filtering, regularization, and related methods can recover accuracy under shift, but the same update can also make predictions worse Wang et al. (2021); Niu et al. (2022; 2023); Wang et al. (2022). An adaptation method proposes a candidate; a deployment system must still decide whether to use it.

Our primary research question is: *When can evidence available without new target labels justify replacing a frozen predictor with a fixed adapted candidate?* Let f_0 be the frozen predictor and f_a the candidate. Their population benefit is

$$\Delta = R_T(f_0) - R_T(f_a). \tag{1}$$

A positive benefit favors the update; a negative benefit favors the frozen model. We distinguish three conclusions: ADAPT supports positive benefit, FREEZE supports negative benefit, and ABSTAIN records insufficient evidence for either strict direction. Both non-adapt actions retain f_0, but only freeze asserts that the candidate is worse.

K-Bound studies this decision as a partial-identification problem. In a binary model, we hold the input law, predictors, and label-free evidence fixed and vary the target labels within a declared class. The resulting feasible benefit set gives an exact boundary for a uniformly justified strict decision. The bound on the hidden calibration residual is an assumption about that class, not a quantity that an unlabeled batch supplies for free. A second result makes this information limit explicit: an audit using the same evidence must still cover the unresolved residual range.

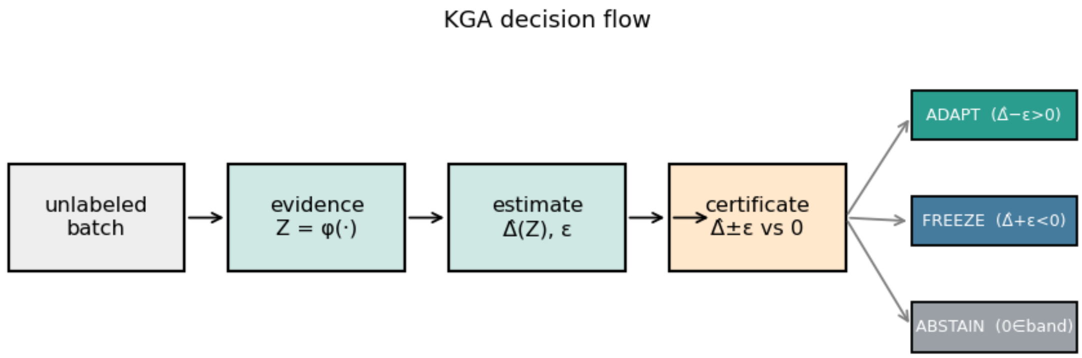

Figure 1: KGA's empirical workflow. A shadow candidate proposes an update; a learned benefit prediction and residual interval determine whether to accept it. Historical labels support development and calibration, while the new decision uses no evaluation labels.

Knowability-Guided Adaptation (KGA) is an *empirical companion* to this analysis. It predicts the observed benefit of a model pair on an evaluation cell and acts only when a residual-based interval excludes zero. KGA is label-free at deployment, not during development and calibration: historical labeled outcomes train the benefit predictor and calibrate its errors. It neither estimates the population residual bound nor implements the population frontier numerically. Section 6.1 explains the common decision logic and the different guarantees.

1.1 Three Scientific Contributions

1. **An exact benefit frontier on a declared class.** We construct target label laws with identical label-free evidence, derive their feasible benefit interval, and characterize its strict positive, strict negative, and unresolved regions, including the zero-benefit boundary.

2. **An evidence-fibre audit limit.** We identify the least residual bound that the same observable law can support uniformly, explaining why historical labels help only through a stated link to the deployment domain.

3. **An empirical paired-benefit gate.** We give a reproducible three-way interval procedure and evaluate its commitment behavior under controlled and natural shifts. The standard coverage-to-action implication states exactly what follows if the interval covers its declared target.

The controlled CIFAR-10-C results show useful routing for some candidates, not a universal improvement: Tent and EATA have lower point regret than either fixed policy, whereas SAR favors always-adapt. The CCT-20 study shows a different behavior: KGA retains frozen predictions and avoids the measured loss from always adapting, but never selects ADAPT. These two studies separate selective routing from conservative retention. The supplementary results retain adverse, incomplete, and withheld studies, so this distinction does not depend on discarding unfavorable evidence.

2 Related Work

2.1 Candidate Adaptation and Update Selection

Tent, EATA, SAR, CoTTA, MEMO, SHOT, and test-time training improve candidate construction through entropy objectives, sample filtering, regularization, temporal stabilization, augmentation, or auxiliary tasks Wang et al. (2021); Niu et al. (2022; 2023); Wang et al. (2022); Zhang et al. (2022); Liang et al. (2020); Sun et al. (2020). Selective methods also choose where or when to update. TTSA selects modality-specific adapters Chen et al. (2025); GALA selects layers and filters samples using gradient alignment Sahoo et al. (2025); Informed Adaptation learns topological features associated with useful updates Mutlu et al. (2026). KGA evaluates a fixed proposed candidate against its frozen baseline. Learned routing and rollback are established ideas; their existence is not our novelty claim.

2.2 Label-Free Performance and Strategy Estimation

ATC, agreement-on-the-line, AETTA, and prediction-matrix statistics estimate performance from unlabeled outputs and suitable development information Garg et al. (2022); Baek et al. (2022); Miller et al. (2021); Lee et al. (2024); Deng et al. (2023); Xie et al. (2024). The closest empirical comparison is Kim et al.'s NeurIPS 2024 work, *Test-Time Adaptation Induces Stronger Accuracy and Agreement-on-the-Line* Kim et al. (2024). It uses agreement-based ALine estimators after adaptation, supports hyperparameter selection, and compares adaptation strategies without labeled OOD data.

KGA instead fits the paired observed benefit of one frozen/candidate pair and uses an interval to make a three-way decision. This is a difference in the fitted target and decision procedure, not a claim that accuracy estimates cannot be differenced to compare models. A current-policy head-to-head comparison with the published TTA/ALine procedure and AETTA remains missing.

2.3 Monitoring and Adaptation Controllers

POEM protects adaptation through entropy matching and online betting Bar et al. (2024). Schirmer et al. monitor the running risk of an adapting model relative to a source-risk reference, under a proxy-informativeness condition Schirmer et al. (2025); the supervised precursor uses a labeled test stream Podkopaev and Ramdas (2022). Drift-to-Action Controllers use delayed labels and online certificates to gate adaptation, abstention, rollback, and retraining Lamaakal et al. (2026). These methods differ in the risk target, label access, and sequential guarantee. KGA concerns one paired benefit and does not establish an anytime-valid risk guarantee.

2.4 Partial Identification and Robust Decisions

An identified set describes the values compatible with the observations and assumptions. Decision theory under partial identification studies how to act when that set leaves payoffs ambiguous; Christensen et al. combine worst-case payoff uncertainty with statistical learning of identified parameters Christensen et al. (2026). K-Bound addresses the narrower question of a uniformly correct strict benefit sign. It does not solve their general optimal-decision problem or claim that interval thresholding itself is new.

Indistinguishable adaptation problems are already central to domain-adaptation impossibility Ben-David et al. (2010a;b). K-Bound's specific contribution is a disagreement-conditional label-kernel construction, its clipped benefit set, and the associated residual-audit floor. The proof does not introduce label-free impossibility. Risk-estimation assumptions based on conditional independence Steinhardt and Liang (2016), label shift Lipton et al. (2018); Garg et al. (2020), or disagreement discrepancy Rosenfeld and Garg (2023) restrict the compatible laws. Likewise, MMD-credal risk intervals impose RKHS and covariate-shift structure Ariq (2026). Such restrictions can change the identified set; they are not contradicted by a converse theorem for a richer class.

2.5 Marginal and Conditional Coverage

Selective classification and reject-option learning abstain on individual predictions Geifman and El-Yaniv (2017); Chow (1970); Cortes et al. (2016). KGA abstains on an update. Its coverage-to-action proposition applies standard interval logic Tibshirani et al. (2019); Barber et al. (2023); Bates et al. (2021); Angelopoulos et al. (2025). The main statistical work is to justify coverage, especially after selection or under shift.

Jin and Ren's JOMI procedure targets coverage conditional on a unit being selected, under exchangeability and a compatible selection rule Jin and Ren (2025). Gibbs et al. provide conditional guarantees over specified classes of shifts using a different calibration construction Gibbs et al. (2025). KGA's ordinary residual interval implements neither procedure. Its marginal false-adapt bound therefore must be distinguished from

Table 1: Closest roles, targets, and remaining distinctions. Literature comparisons are conceptual; the paper does not report a synchronized official head-to-head experiment.

Approach	Main target	Relation to this paper
TTA/ALine; AETTA	adapted-model performance and selection	closest empirical estimators; their scores can inform paired comparisons
POEM; risk monitors	protected adaptation or running risk	different monitored target and sequential assumptions
Partial-identification decisions	uncertain payoffs across compatible laws	context for the population benefit set and robust sign decision
Selection/shift-conditional conformal	coverage after selection or within a shift class	stronger, specialized targets not supplied by KGA's marginal interval
K-Bound / KGA	feasible population benefit / predicted cell benefit	shared update question, distinct uncertainty objects and guarantees

Table 2: Notation and probability targets. Population and measured-cell quantities are kept separate.

Quantity	Level	Meaning
f_0, f_a	both	frozen predictor and realized shadow candidate
Δ	population	expected risk reduction $R_T(f_0) - R_T(f_a)$
$D,\ \mu_T(D)$	population	disagreement region and its target probability
M	population	label-free mean correctness-score margin on D
$\gamma,\ \beta$	population	hidden calibration residual and its externally declared absolute bound
Δ^{cell}	empirical	observed score difference on one predeclared evaluation unit
$\widehat{\Delta},\ \varepsilon$	empirical	predicted cell benefit and residual-based interval radius
FA_u	empirical	probability of ADAPT and a nonpositive cell benefit
FA_c	empirical	probability of nonpositive cell benefit conditional on ADAPT
Commitment rate	empirical	fraction of ADAPT or FREEZE decisions; not interval coverage

error conditional on ADAPT and from coverage within each target environment. Learn Then Test separately addresses configuration selection and multiplicity Angelopoulos et al. (2025).

3 Setup, Notation, and Claim Levels

3.1 Model Pair, Evidence, and Evaluation Unit

Let P_S and P_T be the source and target laws and $R_T(f) = \mathbb{E}_{(X,Y)\sim P_T}[\ell(f(X), Y)]$. Population risks integrate a fresh draw from P_T, holding the realized model pair fixed. An adapter $\mathcal{A}$ proposes f_a from f_0 and unlabeled inputs $X_{1:m}$. The decision can read

$$Z = \phi(X_{1:m}, f_0, f_a),$$

including predictions, logits, and update statistics, but not the new evaluation labels.

For evaluation unit j, with records $\mathcal{E}_j$ and a fixed higher-is-better score S, define

$$\Delta_j^{\text{cell}} = S(f_a; \mathcal{E}_j) - S(f_0; \mathcal{E}_j).$$

The two models are scored on the same records. This measured quantity is not population risk. For accuracy it is a sample average of paired correctness differences; for macro-F1 it is a difference of dataset-level scores, not an average of per-example losses. Metric scales are reported separately.

3.2 Actions and Their Operational Meaning

Definition 1 (Decision semantics)**.** ADAPT commits f_a when the interval supports positive benefit. FREEZE rejects the candidate when it supports negative benefit. ABSTAIN retains f_0 when neither strict direction is supported or the assessment is unavailable.

Freeze is a supported rejection. Abstention is an unresolved assessment. An operator can attach monitoring, label acquisition, or human review to the abstention record; the experiments here evaluate only the frozen

Table 3: Three claim levels and their validity obligations. This table collects the distinctions used throughout the paper.

Claim level	Required premise	Conclusion and boundary
Population frontier	binary 0/1 loss, fixed augmented evidence, declared residual class; richness for necessity	exact strict-sign frontier within that class, not a learned residual bound
Operational proposition	marginal interval coverage for a named scalar target and unit	marginal false commitment for that target, not automatic conditional or anytime control
Empirical evaluation	fixed recorded candidates, splits, metric and action rule	measured cell regret, interval inclusion and action exposure; population transfer needs a separate argument

fallback and uncertainty log, not those additional interventions. Retaining f_0 after missing evidence must be logged as ABSTAIN, not as a certified FREEZE.

The observed-cell oracle chooses the larger score. Regret is the policy's score shortfall from that oracle, averaged over the named units. Always-freeze regret is $\max(\Delta_j^{\text{cell}}, 0)$; always-adapt regret is $\max(-\Delta_j^{\text{cell}}, 0)$. KGA uses the candidate score only after ADAPT. Lower regret is better, and a zero-benefit tie carries zero regret for either model even though it supports neither strict sign.

3.3 Commitment, Interval Coverage, and Error

These are three different measurements:

$$\text{Commit} = \mathbb{P}(g \in \{\textsc{adapt}, \textsc{freeze}\}), \tag{2}$$

$$\text{Cov}_{\text{cell}} = \mathbb{P}(|\widehat{\Delta} - \Delta^{\text{cell}}| \le \varepsilon), \tag{3}$$

$$\text{FA}_{\text{u}} = \mathbb{P}(g = \textsc{adapt}, \Delta^{\text{cell}} \le 0). \tag{4}$$

Conditional false adaptation is $\text{FA}_{\text{c}} = \mathbb{P}(\Delta^{\text{cell}} \le 0 \mid g = \textsc{adapt})$, defined only when $\mathbb{P}(g = \textsc{adapt}) > 0$. The operational proposition below bounds the marginal error under interval coverage; it does not bound FA_{c}. A gate that never adapts has zero false adaptations mechanically, so every error report includes action exposure. Analogously, a false FREEZE is a FREEZE decision with $\Delta^{\text{cell}} \ge 0$; its conditional frequency divides by FREEZE decisions and is undefined when that action is absent.

The population theorem uses binary labels and 0/1 loss. The empirical gate can instead target any predeclared scalar score difference, including multiclass metrics. This interface does not extend the binary theorem. In general multiclass classification, $\Delta = \mathbb{P}_T(D)(p_a - p_0)$, with $p_a = \mathbb{P}_T(f_a = Y \mid D)$ and $p_0 = \mathbb{P}_T(f_0 = Y \mid D)$; only the binary setting forces $p_0 = 1 - p_a$.

4 Population Partial-Identification Frontier

The result describes which benefit signs are compatible with a fixed label-free evidence law and a declared bound on the hidden residual.

4.1 Disagreement-Region Model

Assumption 1 (Deployment setup)**.** Let $Y \in \{0, 1\}$, use 0/1 loss, fix measurable binary predictors f_0, f_a, and assume $\mu_T(D) > 0$ for $D = \{x : f_0(x) \neq f_a(x)\}$. Let $s : \mathcal{X} \to [0, 1]$ be a fixed measurable candidate-correctness score, fitted using source information, and let the target law admit the measurable pointwise correctness kernel $\eta_a(x) = \mathbb{P}_T(f_a(X) = Y \mid X = x)$. On D, set

$$M = \mathbb{E}[s(X) \mid D] - \tfrac{1}{2}, \qquad \gamma = \mathbb{E}[\eta_a(X) - s(X) \mid D]. \tag{5}$$

The declared population class is $\mathcal{C}_\beta = \{P_T : |\gamma(P_T)| \le \beta\}$ for an externally supplied $\beta \ge 0$. Because $s \in [0, 1]$, the feasible margin satisfies $M \in [-\frac{1}{2}, \frac{1}{2}]$. The identities and frontier below do not require a source-calibration property of s.

Remark 1 (Residual identity). The definition gives

$$\begin{aligned} M+\gamma &= \left(\mathbb{E}[s(X) \mid D]-\tfrac{1}{2}\right)+\mathbb{E}[\eta_a(X)-s(X) \mid D] \\ &= \mathbb{E}[\eta_a(X) \mid D]-\tfrac{1}{2}. \end{aligned}$$

This is an identity, not empirical evidence that calibration transfers. We call γ a target disagreement-conditional calibration residual, not automatically source-to-target drift. Ordinary source calibration need not hold after conditioning on D, so γ can be nonzero even when $P_T = P_S$. A literal label-kernel shift interpretation would require an additional reference, for example the pointwise source correctness $s(x) = \mathbb{P}_S(f_a(X) = Y \mid X = x)$. The no-shift counterexample is given in Appendix L. Calibration notions also depend on the conditioning information: top-label calibration, for example, additionally conditions on the predicted class Gupta and Ramdas (2022). Neither score nor top-label calibration automatically supplies calibration conditional on D.

Definition 2 (Risk alignment)**.** Evidence Z is risk-aligned on a class $\mathcal{P}$ if no two laws in $\mathcal{P}$ with opposite nonzero benefit signs induce the same law of Z. This is weaker than uniform strict-sign identification: a fibre containing zero and positive benefit is risk-aligned by this definition, but does not support a uniformly positive claim. Population frontier claims take M to be fixed on each augmented evidence fibre, equivalently by using $\widetilde{Z} = (Z, M)$.

Definition 3 (Strict directional soundness)**.** At a fixed augmented-evidence law, ADAPT is uniformly directionally sound when $\Delta(P) > 0$ for every compatible $P \in \mathcal{C}_\beta$; FREEZE is sound when $\Delta(P) < 0$ for every such law. A pointwise maximal three-way rule commits whenever one strict action is sound and abstains otherwise. The strict convention treats $\Delta = 0$ as insufficient for either directional claim, although the actions then have equal task risk.

4.2 Label-Kernel Freedom on the Disagreement Region

Lemma 1 (Label-kernel freedom on disagreement)**.** *Fix $(\mu_T, P_S, f_0, f_a, s)$, hence D and M, and fix the conditional label law outside D. For every measurable $\eta : D \to [0,1]$ there is a target law with input marginal μ_T and $\mathbb{P}(f_a(X) = Y \mid X = x) = \eta(x)$ on D. Every label-free observable formed from the input batch, the fixed models, and their outputs has the same law for all such η, while*

$$\gamma = \mathbb{E}[\eta(X) \mid D] - \tfrac{1}{2} - M.$$

Proof. On D, binary predictors disagree, so their outputs are 0 and 1. Assign conditional mass $\eta(x)$ to $Y = f_a(x)$ and $1 - \eta(x)$ to $Y = f_0(x)$, retaining the fixed kernel outside D. This changes no input marginal or model output and therefore changes no label-free evidence law. The expression for γ follows from Eq. (5). □

4.3 Benefit Reduction and Matched-Evidence Impossibility

Lemma 2 (Disagreement-region reduction)**.** *Under Assumption 1, with $\bar{a} := \mathbb{P}_T(f_a{=}Y \mid D)$,*

$$\begin{aligned} \Delta &= 2\,\mu_T(D)\left(\bar{a}-\tfrac{1}{2}\right), \\ \operatorname{sign}\Delta &= \operatorname{sign}(\bar{a}-\tfrac{1}{2}) = \operatorname{sign}(M+\gamma). \end{aligned}$$

The identity $\operatorname{sign}\Delta = \operatorname{sign}(M+\gamma)$ *holds for* every *target law; it does not assume $|\gamma| \le \beta$.*

Proof. On D exactly one of f_0, f_a is correct under 0/1 loss, so the benefit equals $2\mu_T(D)(\bar{a} - \frac{1}{2})$. By the definitions of M and γ,

$$\begin{aligned} M+\gamma &= \left(\mathbb{E}_{\mu_T|D}[s]-\tfrac{1}{2}\right)+\mathbb{E}_{\mu_T|D}[\eta_a - s] \\ &= \mathbb{E}_{\mu_T|D}[\eta_a]-\tfrac{1}{2} = \bar{a}-\tfrac{1}{2}, \end{aligned}$$

so $\operatorname{sign}\Delta = \operatorname{sign}(\bar{a}-\frac{1}{2}) = \operatorname{sign}(M+\gamma)$. □

Theorem 1 (Interior matched-evidence impossibility)**.** *Fix $\beta > 0$. For any feasible margin $M \in [-\frac{1}{2}, \frac{1}{2}]$ with $|M| < \beta$ there exist $P_T^1, P_T^2 \in \mathcal{C}_\beta$ such that $\mathrm{Law}(\widetilde{Z} \mid P_T^1) = \mathrm{Law}(\widetilde{Z} \mid P_T^2)$ for $\widetilde{Z} = (Z, M)$, $M(P_T^1) = M(P_T^2) = M$, and $\operatorname{sign} \Delta_1 = -\operatorname{sign} \Delta_2 \neq 0$. (For $\beta = 0$ the class $\mathcal{C}_0$ forces $\gamma = 0$, so $\operatorname{sign} \Delta = \operatorname{sign} M$ is determined and no such pair exists: matched-evidence ambiguity requires a positive declared residual bound.)*

Proof. This is Lemma 1 read at one radius. Choose $0 < \delta < \min\{\beta - |M|, \frac{1}{2}\}$ and apply the lemma with the two constant kernels $\eta \equiv \frac{1}{2} + \theta\delta$, $\theta \in \{+1, -1\}$, holding the off-D kernel K° common to both worlds. The lemma supplies the two laws P_T^θ, and: *class membership,* $\bar{a}_\theta = \frac{1}{2} + \theta\delta$ so $\gamma_\theta = \theta\delta - M$ with $|\gamma_\theta| \leq \delta + |M| < \beta$, hence both worlds lie in $\mathcal{C}_\beta$; *matched evidence,* $\mathrm{Law}(W \mid P_T^\eta)$ does not depend on η by the lemma and $\widetilde{Z} = (Z, M)$ is a function of W, so $\mathrm{Law}(\widetilde{Z} \mid P_T^+) = \mathrm{Law}(\widetilde{Z} \mid P_T^-)$; *opposite signs,* Lemma 2 gives $\Delta_\theta = 2\mu_T(D)\,\theta\delta \neq 0$ with $\operatorname{sign} \Delta_\theta = \theta$. ☐

The two evidence laws here coincide *exactly*, not merely approximately: the construction changes only the label kernel on D, and Z is a function of unlabeled inputs and model outputs. We therefore do not present this as a Le Cam two-point argument. A Le Cam argument trades off a positive total-variation distance against the achievable error; here the distance is zero, which is the degenerate and strongest case, and the corresponding statement is exact non-identifiability rather than a rate.

Corollary 1 (Matched-evidence abstention lower bound)**.** *Fix $\alpha \in (0, \frac{1}{2})$. On the shared evidence law constructed in Theorem 1, any possibly randomized label-free rule g satisfying $\mathbb{P}[g = \textsc{adapt}, \Delta \leq 0] \leq \alpha$ in every admissible world and $\mathbb{P}[g = \textsc{freeze}, \Delta \geq 0] \leq \alpha$ in every admissible world must satisfy $\mathbb{P}[g = \textsc{abstain}] \geq 1 - 2\alpha$.*

Proof. The rule has the same action probabilities in the two evidence-identical worlds. Its adapt probability is at most α in the negative world and its freeze probability is at most α in the positive world. The three action probabilities sum to one. ☐

4.4 Boundary Case

Proposition 1 (Boundary case and closed-band abstention)**.** *Fix $\alpha \in (0, \frac{1}{2})$. Under Definition 3, abstention is the pointwise maximal sound three-way action throughout the closed band $|M| \leq \beta$, and any label-free rule controlling both marginal errors at level α abstains there with probability at least $1 - 2\alpha$. Opposite nonzero signs are asserted only in the interior $|M| < \beta$.*

Proof. For $|M| < \beta$ this is Theorem 1 together with Corollary 1. For $|M| = \beta > 0$, the declared class contains an augmented-evidence-identical zero-benefit world (take $\gamma = -M$, admissible since $|\gamma| = \beta$) and a strict-benefit world of sign M (take $\gamma = 0$). In the zero-benefit world $\Delta = 0$, so *both* directional error events $\{\textsc{adapt}, \Delta \leq 0\}$ and $\{\textsc{freeze}, \Delta \geq 0\}$ occur whenever the rule commits; since the rule's action probabilities are identical across the two evidence-identical worlds, dual control at level α gives $\mathbb{P}[\textsc{adapt}] \leq \alpha$ and $\mathbb{P}[\textsc{freeze}] \leq \alpha$, hence $\mathbb{P}[\textsc{abstain}] \geq 1 - 2\alpha$, and neither strict action is uniformly directionally sound. For $M = \beta = 0$ the class $\mathcal{C}_0$ forces $\gamma = 0$, so $\Delta \equiv 0$ and the same conclusion holds directly. ☐

This directional convention is stricter than task regret: at $\Delta = 0$, adapt and freeze are risk-equivalent but neither certifies a strict sign.

4.5 Exact Strict-Commitment Frontier

Theorem 2 (Exact strict-commitment frontier)**.** *Under Assumption 1, fix the observables $(\mu_T, P_S, f_0, f_a, s)$ so M is determined, and let $\beta \geq 0$. The necessity and maximality clauses use the full class $\mathcal{C}_\beta$ and therefore its richness: it contains the evidence-identical label kernels constructed in Theorem 1 and the zero-benefit boundary law. For a narrower declared subclass not closed under these constructions, only the sufficiency clause follows without an additional richness assumption.*

(i) ***Reduction.*** *For every P_T, $\operatorname{sign} \Delta = \operatorname{sign}(M + \gamma)$ (Lemma 2).*

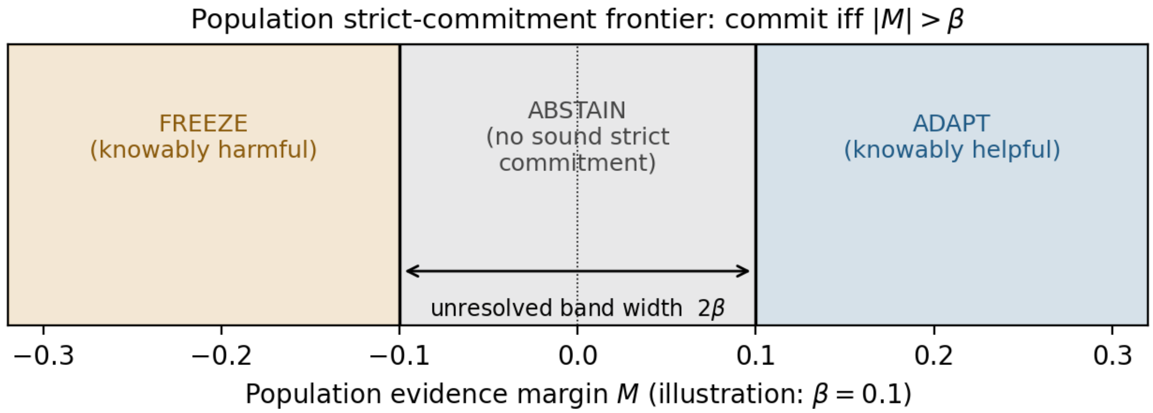

Figure 2: Population strict-commitment frontier. $M > \beta$ supports adapt; $M < -\beta$ supports freeze; and $|M| \leq \beta$ supports no uniform strict commitment. Interior ambiguity uses opposite nonzero benefit signs, whereas the boundary uses zero-benefit ambiguity. This population rule is not the numerical KGA rule used on the real-data panels.

(ii) ***Sufficiency.*** *If* $|M| > \beta$ *and* $P_T \in \mathcal{C}_\beta$, *then* $\operatorname{sign}\Delta = \operatorname{sign} M$.

(iii) ***Necessity.*** *If* $|M| < \beta$ *(which requires* $\beta > 0$*), there exist* $P_T^1, P_T^2 \in \mathcal{C}_\beta$ *with the same augmented-evidence law and opposite* nonzero *benefit signs; at* $|M| = \beta > 0$ *the admissible residual* $\gamma = -\operatorname{sign}(M)\beta$ *yields* $\Delta = 0$, *and in the degenerate case* $M = \beta = 0$ *the class* $\mathcal{C}_0$ *forces* $\Delta \equiv 0$. *Hence* no strict *benefit direction is uniformly supported by* $\operatorname{Law}(\widetilde{Z})$ *over* $\mathcal{C}_\beta$ *whenever* $|M| \leq \beta$.

(iv) ***Strict-commitment iff.*** *At the fixed augmented-evidence law, a strict action in* {ADAPT, FREEZE} *is uniformly directionally sound in the sense of Definition 3 if and only if* $|M| > \beta$; *on the closed band* $|M| \leq \beta$ *(including* $M = \beta = 0$, *where* $\Delta \equiv 0$*) abstention is the pointwise maximal sound action.*

The population rule $M > \beta \Rightarrow$ ADAPT, $M < -\beta \Rightarrow$ FREEZE, $|M| \leq \beta \Rightarrow$ ABSTAIN *is the pointwise maximal sound certificate on* $\mathcal{C}_\beta$.

Proof. Part (i) is Lemma 2. For (ii), if $M > \beta$ and $|\gamma| \leq \beta$, then $M + \gamma > 0$, so $\operatorname{sign}\Delta = \operatorname{sign}(M + \gamma) = +1$; the negative case is symmetric. For (iii), Theorem 1 supplies two laws in the same augmented-evidence fibre with opposite nonzero signs whenever $|M| < \beta$. At $|M| = \beta$ the boundary law with $\gamma = -\operatorname{sign}(M)\beta$ gives $\Delta = 0$, still precluding a determined nonzero sign. Part (iv) combines (ii), (iii), and Proposition 1. Pointwise maximality follows because (ii) certifies the unique strict direction outside the band, while Theorem 1 and Proposition 1 rule out both strict directions inside and on the boundary. ☐

The feasible range can also be read directly from the identified benefit set. Write $d = \mu_T(D) > 0$. Over the full declared class with the observables fixed,

$$\Delta(\mathcal{C}_\beta) = 2d\left[\max\{-\tfrac{1}{2}, M - \beta\}, \min\{\tfrac{1}{2}, M + \beta\}\right].$$

Indeed, $\bar{a} - \frac{1}{2} \in [-\frac{1}{2}, \frac{1}{2}]$ and $|\bar{a} - \frac{1}{2} - M| \leq \beta$; every value in their intersection is attained by a constant correctness kernel on D. A strict direction is available exactly when this whole interval lies above or below zero.

Region	Meaning	Action
$M > \beta$	strictly positive benefit	ADAPT
$M < -\beta$	strictly negative benefit	FREEZE
$\lvert M\rvert < \beta$	both strict signs are feasible	ABSTAIN
$\lvert M\rvert = \beta > 0$	zero benefit is feasible	ABSTAIN
$M = \beta = 0$	benefit is identically zero	ABSTAIN

4.6 Interpretation and Limits of the Frontier

The theorem depends on a declared class. Choosing $\beta = 0$ assumes a zero target disagreement-conditional residual; it is not a conservative default. Without a credible upper bound on $|\gamma|$, the population frontier cannot support a robust strict action. More unlabeled samples do not solve this problem because the missing difference lies in the target labels, not in the input batch.

Empirical abstention is not automatically evidence of structural unknowability. A wide finite-sample interval may instead reflect limited calibration data, estimator error, or transfer failure. The frontier therefore leaves one central question: can its required residual bound be learned from the same label-free evidence?

5 What Label-Free Evidence Cannot Learn

5.1 Audit Setup and Evidence Fibres

Let W collect everything a deployment-time audit may read: the unlabeled target batch, the fixed models and score function, label-free statistics derived from them, and any source or calibration sample whose law is specified independently of the target label kernel. An audit is a nonnegative random quantity $\widehat{\beta} = g(W, U)$, where U is independent randomization.

For a declared class $\mathcal{C}$ and an observable law $z = \mathrm{Law}(W)$, define the evidence fibre

$$\mathcal{C}(z) = \{P \in \mathcal{C} : \mathrm{Law}(W \mid P) = z\}$$

and its unresolved residual radius

$$\Gamma_z(\mathcal{C}) = \sup_{P \in \mathcal{C}(z)} |\gamma(P)|. \tag{6}$$

The fibre collects target laws that the audit cannot distinguish. Its radius is therefore the smallest uniform residual bound that those observations could support.

5.2 Exact Fibre-Wise Audit Floor

Theorem 3 (Exact label-free audit floor). *Fix a nonempty fibre $\mathcal{C}(z)$ and $\delta \in [0, 1)$. If $\widehat{\beta}$ satisfies $\mathbb{P}_P(|\gamma(P)| \leq \widehat{\beta}) \geq 1 - \delta$ for every $P \in \mathcal{C}(z)$, then*

$$\mathbb{P}\Big(\widehat{\beta} \geq \Gamma_z(\mathcal{C})\Big) \geq 1 - \delta.$$

The constant audit $\widehat{\beta} \equiv \Gamma_z(\mathcal{C})$ is valid with $\delta = 0$. Consequently, on the high-probability event above, an audited frontier commits only where the frontier using the declared radius $\Gamma_z(\mathcal{C})$ already commits.

Proof. Every law in the fibre induces the same law for W, and hence the same law for $\widehat{\beta}$. Validity for each fixed P requires that common law to exceed $|\gamma(P)|$ with probability at least $1 - \delta$. Taking an increasing sequence of attainable absolute residuals approaching the supremum gives the lower bound. The constant audit is valid by the definition of the supremum. If $\widehat{\beta} \geq \Gamma_z$, then $|M| > \widehat{\beta}$ implies $|M| > \Gamma_z$. □

This result does not say that useful budgets never exist. It says that the same label-free fibre cannot create information about the target label kernel that the fibre definition removes. A valid audit must cover the unresolved radius; reading more functions of the same W does not shrink it. The constant $\Gamma_z(\mathcal{C})$ is an oracle benchmark for the fixed evidence-law fibre. The theorem does not provide an algorithm that learns the unknown law z or its radius from a finite observed batch. We therefore describe this as a fibre-wise audit floor rather than an unspecified minimax estimation result.

5.3 Declared Budget and Abstention Radius

For the full declared class $\mathcal{C}_\beta = \{P : |\gamma(P)| \leq \beta\}$ with $0 \leq \beta \leq \frac{1}{2}$, the label-kernel construction in Lemma 1 gives

$$\Gamma_z(\mathcal{C}_\beta) = \beta. \tag{7}$$

Thus the least fibre-wide budget a label-free audit can certify is the budget already declared by the class. The same number is also the population abstention radius:

$$\boxed{\begin{array}{c}\text{least valid fibre-wide residual bound} \\ = \text{population abstention radius} = \beta\end{array}}$$

This equality connects the audit problem to the strict-commitment frontier. It does not turn the declaration into an estimate.

5.4 Why Labels at the Wrong Domain Buy Nothing

Suppose the audit also observes a fully labeled calibration domain, but the declared class contains no coupling between that domain's label kernel and the deployment label kernel. The construction in Lemma 1 changes only deployment labels on D and leaves the calibration sample unchanged. The augmented evidence law is therefore identical across the same target fibre, so Theorem 3 applies without a smaller radius. Labels help only when they come from the deployment domain or when a stated assumption links the labeled and deployment domains.

5.5 What Additional Structure Can Buy

The missing information can be supplied, but each route changes the problem. Deployment labels can estimate target correctness directly. Historical labeled deployments can support an episode-marginal budget under an exchangeability model. A structural coupling can bound the change between calibration and target domains. Domain knowledge can declare a physically meaningful limit. These are purchases made with labels or assumptions; they are not consequences of the current unlabeled batch. Detailed rate results and extended constructions belong to the long-form theory record and are not required by the empirical claims here.

5.6 Implications for K-Bound

KGA does not estimate β and does not numerically apply $|M| > \beta$ on the reported real-data benchmarks. Its practical certificate instead places a finite-sample uncertainty radius ε around the predicted cell benefit $\widehat{\Delta}$. Population benefit Δ and observed cell benefit Δ^{cell} concern the same model pair at different levels. Transferring coverage from the cell outcome to population risk requires a separate sampling-error bound; the reported empirical intervals do not establish that bound.

6 KGA: An Empirical Benefit-Interval Gate

6.1 What Transfers from the Population Analysis

KGA is an empirical companion to K-Bound, not an implementation of its population frontier. Both ask whether to accept the same proposed update and both leave unresolved evidence as abstention. Their uncertainty objects differ. The binary population model gives

$$\Delta = 2\mu_T(D)(M + \gamma). \tag{8}$$

The observed cell outcome is

$$\Delta^{\text{cell}} = S(f_a; \mathcal{E}) - S(f_0; \mathcal{E}),$$

and the fitted empirical benefit prediction is

$$\widehat{\Delta} = h_\theta(Z).$$

KGA does not estimate M, γ, or β, and does not numerically apply $|M| > \beta$. Its radius ε describes prediction error for a measured cell outcome, not the range of population benefits compatible with a label-free evidence law.

KGA is label-free at deployment, not during development and calibration. Historical labeled cell outcomes fit h_θ and supply residuals. At the new decision, KGA reads only Z and the frozen fitted artifacts. The interval premise for the empirical target is

$$\mathbb{P}(|\widehat{\Delta} - \Delta^{\text{cell}}| \le \varepsilon) \ge 1 - \alpha. \tag{9}$$

The proposition below concerns this target when this premise holds.

6.2 The Coverage-to-Action Proposition

The following standard implication makes the decision semantics explicit. It is not a new calibration theorem: coverage is the premise, and establishing it is the statistical burden.

Proposition 2 (Coverage-to-action implication)**.** *Fix the scalar benefit target and its evaluation unit before calibration. Let B and $\widehat{B}$ be finite real random variables, and let $\varepsilon \in [0, \infty]$ be a measurable random radius on a common probability space containing all calibration, benefit-model fitting, candidate-construction, and evaluation-unit randomness. The prediction $\widehat{B}$ and radius ε must be available without reading the new unit's evaluation labels. For $\alpha \in (0, 1)$, suppose the interval satisfies marginal coverage*

$$\mathbb{P}\big[\, |\widehat{B} - B| \le \varepsilon \,\big] \;\ge\; 1 - \alpha.$$

Define

$$g = \begin{cases} \textsc{adapt} & \widehat{B} - \varepsilon > 0, \\ \textsc{freeze} & \widehat{B} + \varepsilon < 0, \\ \textsc{abstain} & \textit{otherwise}. \end{cases}$$

Then, marginally over that common probability space,

$$\mathbb{P}\big[g = \textsc{adapt}, B \le 0\big] \le \alpha,$$
$$\mathbb{P}\big[g = \textsc{freeze}, B \ge 0\big] \le \alpha.$$

Proof. On the event $\{|\widehat{B} - B| \le \varepsilon\}$, the condition $\widehat{B} - \varepsilon > 0$ implies $B > 0$, so a false adapt requires the complement of this event, which has probability at most α. The freeze case is symmetric. □

For the empirical panels, $B = \Delta^{\text{cell}}$ is the observed evaluation-unit score difference and $\widehat{B} = \widehat{\Delta}$. A population-risk application instead takes $B = \Delta$ and requires coverage for that quantity; cell-outcome coverage alone does not supply it. Coverage must be justified separately. Joint exchangeability of calibration and next-unit scores under the fitting and candidate protocol, or a justified shift correction, can support this premise. Risk alignment is not required for the conditional-on-coverage implication.

Remark 2 (Marginal vs. conditional error). Proposition 2 bounds the *marginal, unconditional* false-commit probabilities for the declared target B. Its false-adapt rate is $\mathrm{FA_u} = \mathbb{P}[g = \textsc{adapt}, B \le 0]$. It does *not* bound $\mathrm{FA_c} = \mathbb{P}[B \le 0 \mid g = \textsc{adapt}]$, defined when $\mathbb{P}[g = \textsc{adapt}] > 0$ and reported descriptively for $B = \Delta^{\text{cell}}$. Selection-conditional coverage requires a separate construction, such as that of Jin and Ren Jin and Ren (2025); conditional guarantees over specified shift classes also require their own calibration procedure Gibbs et al. (2025). Neither is supplied by the marginal premise above. Simultaneous protection across candidates and time-uniform protection likewise require additional guarantees. Risk alignment (Definition 2) concerns population sign identifiability; coverage for B alone yields the *interval* implication above.

Remark 3 (Fallback when assumptions are unsupported). If interval coverage for the declared target and evaluation unit—or the assumptions supporting it—cannot be justified, Proposition 2 supplies no level-α protection for either strict action. A separately declared operational fallback may retain the frozen model and record abstention. Serving the frozen model without a valid negative interval is not a certified FREEZE decision. This policy recommendation does not establish that the retained model is safe or optimal. If *risk alignment* (Definition 2) cannot be justified for an intended population sign claim, that interpretation remains unsupported. The interval implication of Proposition 2 still holds when coverage for the declared B is established (cf. Theorem 1).

Algorithm 1 KGA for one proposed update

Require: Frozen f_0, adapter $\mathcal{A}$, unlabeled X, frozen h_θ and calibration artifacts

Create $f_a \leftarrow \mathcal{A}(f_0, X)$ in isolated shadow state
Extract $Z = \phi(X, f_0, f_a)$ and validate the assessment inputs
if the assessment is unavailable **then**
  retain f_0, log the reason, and return ABSTAIN
end if
Obtain ε from the calibrated residual rule in Eq. (10)
$\widehat{\Delta} \leftarrow h_\theta(Z)$; $L \leftarrow \widehat{\Delta} - \varepsilon$; $U \leftarrow \widehat{\Delta} + \varepsilon$
if $L > 0$ **then**
  commit f_a and return ADAPT
else if $U < 0$ **then**
  discard f_a and return FREEZE
else
  retain f_0, log uncertainty, and return ABSTAIN
end if

A population application would instead require coverage of Δ. For accuracy, one possible bridge adds a bound b for $|\Delta^{\text{cell}} - \Delta|$ to the empirical radius; the triangle inequality and union bound then give a population interval with radius $\varepsilon + b$. The precise conditional statement is in Appendix A. No such sampling-error bound is established for the reported panels. In particular, when a candidate adapts to evaluation inputs, hiding their labels does not make the candidate independent of those inputs.

6.3 Evidence, Calibration, and the Action Rule

The controlled implementation stores 11 label-free features, with effective rank 9 because two features are deterministic differences. They summarize confidence, class balance, model disagreement, and update size. For each dataset and adapter, a separate `GradientBoostingRegressor` Pedregosa et al. (2011) uses squared-error loss, 250 trees, depth 2, learning rate 0.05, subsample 0.8, and seed 0. Natural protocols retain their own fitted model and feature schema; for example, CCT-20 uses a ridge model. Details are in Appendix I. A common interface does not imply that an estimator or radius transfers across adapters.

Given residuals $r_i = |\widehat{\Delta}_i - \Delta_i^{\text{cell}}|$, ordered as $r_{(1)} \leq \cdots \leq r_{(n)}$, the order-statistic rule is

$$k = \lceil (n+1)(1-\alpha) \rceil, \qquad \varepsilon = \begin{cases} r_{(k)}, & k \leq n, \\ +\infty, & k > n. \end{cases} \tag{10}$$

Split-conformal coverage requires joint exchangeability of the calibration and next-unit scores under the fitting and candidate-construction protocol Tibshirani et al. (2019); Barber et al. (2023). An independently fitted predictor and exchangeable calibration/deployment units are one sufficient route. The rank counts evaluation units, not the images inside them.

The controlled grids use *leave-one-cell-out empirical order-statistic calibration*. Their benefit predictions are cross-fitted, and each scored residual is excluded from its radius. Related cells still share corruption families, checkpoints, and residual pools, so this is an empirical procedure, not exact split conformal or jackknife+. Natural protocols use separated development, calibration, and target partitions where the archived design records that separation.

For example, prediction 0.04 and radius 0.01 give $[0.03, 0.05]$ and ADAPT; prediction -0.04 gives $[-0.05, -0.03]$ and FREEZE; prediction 0.005 gives $[-0.005, 0.015]$ and ABSTAIN. These illustrate the rule, not measured results.

Candidate adaptation, frozen and candidate inference, evidence extraction, and shadow-state storage all contribute to runtime. The tree prediction and interval comparison are inexpensive, but the whole procedure

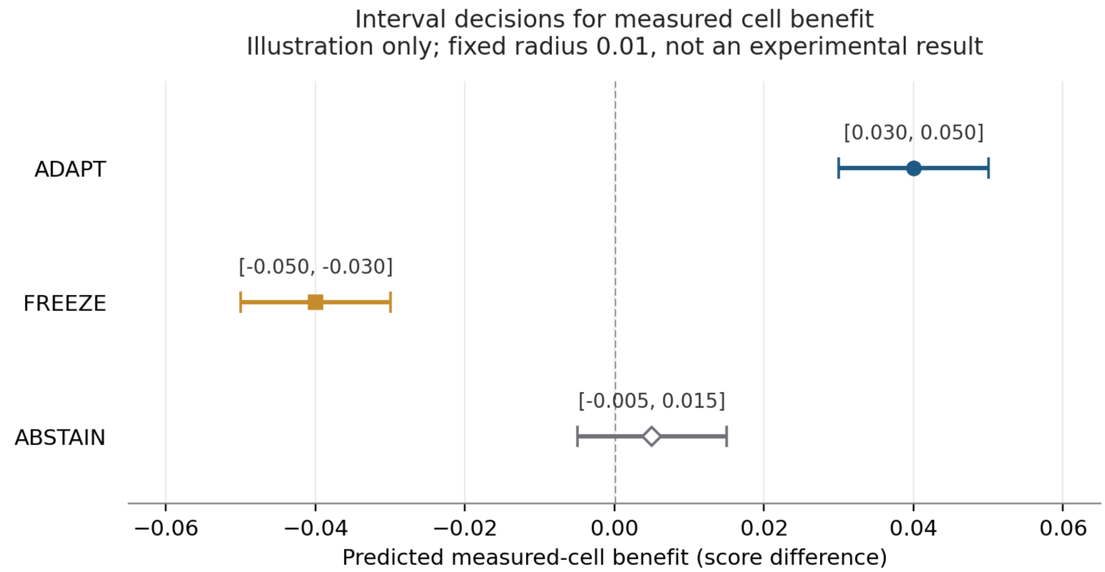

Figure 3: Illustration of the three interval decisions for measured cell benefit. Only an interval strictly above or below zero supports a commitment. The figure illustrates the rule; coverage must be established for the intended evaluation unit.

Table 4: Experiment status and role. Primary presentation does not imply prospective confirmation. All recorded unfavorable outcomes remain disclosed.

Evidence	Design and status	Question it can answer
CIFAR-10-C: Tent, EATA, SAR	opened controlled grids; cross-fitted empirical calibration	candidate-specific routing on the evaluated cells
CCT-20: Tent	internally sealed protocol; five checkpoints and nine target camera locations	conservative retention under the stated score and safe-utility rule
So2Sat-LCZ42	development feasibility stopped; no target access	whether a candidate met the pre-target eligibility checks
Other replayable tracks	opened natural/semantic and candidate-configuration diagnostics	transfer limits and one-sided retention; full results in the supplement
PACS; iWildCam; historical ports; focused Office-Home route	partial replay, withheld metric, superseded policy, or invalid route	audit history only where the corresponding contract is incomplete or invalid

is not one extra forward pass. Comparable end-to-end latency measurements are not available across the recorded tracks.

7 Primary Experimental Protocols

7.1 Two Primary Questions and the Evidence Hierarchy

The controlled study asks whether the gate distinguishes useful from harmful updates. The CCT-20 study asks whether a frozen gate avoids measured candidate harm under a natural camera-location shift. These are different claims. We predefine the interpretation of each study before presenting its results in Table 4.

CIFAR-10-C Hendrycks and Dietterich (2019) supplies 2,160 recorded cells per candidate: five run seeds over 432 corruption/composition/update conditions. The current gate is replayed on the stored benefit predictions and cell outcomes. The analysis uses leave-one-condition-out cross-fitted empirical residual calibration; it is not an untouched environment test. Five run seeds here do not represent five independently trained source checkpoints.

7.2 CCT-20 Design and Safe Utility

The CCT-20 camera-location study Beery et al. (2018) crosses 5 independent source checkpoints with 9 target locations, giving 45 checkpoint–location cells. It uses 6950 probe images and 16325 evaluation images (23275 total). The primary metric is set-membership top-1 accuracy: a prediction is correct if it matches any distinct archived category for that image. It is not an official single-label leaderboard reproduction. The candidate uses the official Tent implementation, one non-episodic Adam step at learning rate 10^{-3}, and state reset per cell.

The protocol was internally sealed before source training and outcome access, not publicly registered. Aggregate target label metadata had been inspected during candidate ranking; cell-level outcomes remained unopened until one-shot scoring. This is outcome-blind execution, not a label-unopened study. The source ridge fit used 10 rows from two camera locations; calibration used 45 rows aggregated to nine cis-test locations. Transfer to trans-test locations remains an assumption.

For either fixed policy b, define G_b as mean baseline regret minus KGA regret. Positive G_b means lower regret for KGA. The *safe-utility* rule is a predeclared, limited harm-avoidance check. Using nominal pointwise 95% bootstrap lower endpoints, it passes when

$$L_{\text{adapt}} > 0, \qquad L_{\text{freeze}} > -0.005. \tag{11}$$

The second threshold is the frozen-policy noninferiority margin, on the accuracy-proportion scale. This rule can pass when KGA always retains f_0. The stronger routing criterion additionally requires improvement over both fixed policies at the declared inference level and nontrivial ADAPT/FREEZE exposure. The word *safe* names this protocol endpoint, not a population guarantee.

7.3 Scoring, Dependence, and Statistical Inference

All policies use the same evaluation records, and evaluation labels are joined only after actions are fixed. The candidate TTA step can be transductive: episodic candidates update on the unlabeled evaluation inputs, and online candidates may use evaluation-batch BatchNorm statistics. Development/calibration separation therefore does not imply independent fresh-sample inference.

We report oracle regret, action counts, commitment rate, and false-commit frequencies. Empirical interval inclusion and interval width are reported separately from commitment. A paired regret contrast always means baseline regret minus KGA regret, so positive values favor KGA. Accuracy and macro-F1 regrets are separated rather than pooled.

Bootstrap levels are nominal throughout. CIFAR sensitivity resamples six corruption families, conditional on an archived checkpoint. CCT-20 uses a paired two-way product bootstrap over checkpoint rows and location columns; sign-flip calculations instead use nine location means. For a vector G of cluster gaps, the sign-flip reference model is

$$G \overset{d}{=} s \odot G \quad \text{for every } s \in \{-1, +1\}^m. \tag{12}$$

Independent zero-symmetric cluster gaps suffice. Exchangeability, marginal symmetry, or a single global sign symmetry alone is not sufficient Hemerik and Goeman (2018). Enumeration removes Monte Carlo error but does not verify this null model. Holm adjustment requires valid input p-values; nominal family intervals also require valid marginal coverage of their constituent intervals. The full inference specification and retrospective search status are collected in Appendix F.

8 Primary Results

8.1 Controlled Routing Depends on the Candidate

Table 5 reports all three CIFAR-10-C candidates under the same current rule. Tent and EATA have lower measured point regret than always-adapt and always-freeze. SAR is the counterexample: always-adapt has lower regret than KGA. The mixed panel therefore supports candidate-specific routing, not a claim that gating improves every adapter.

Table 5: Primary controlled CIFAR-10-C results. Regret is the mean accuracy shortfall from the per-cell oracle, on the proportion scale; lower is better. A/F/U denotes ADAPT/FREEZE/ABSTAIN. Commitment rate is $(A+F)/n$, not interval coverage. Cells share checkpoints and residual pools.

Candidate	n	KGA	Adapt	Freeze	A/F/U	$\mathrm{FA_u}$	Commitment rate
Tent	2160	0.0016	0.0080	0.1239	1107/359/694	0.0000	0.6787
EATA	2160	0.0013	0.0033	0.1313	1241/132/787	0.0000	0.6356
SAR	2160	0.0016	0.0003	0.1405	1446/0/714	0.0000	0.6694

Table 6: Retrospective interval diagnostics for the controlled cells, at nominal target $1-\alpha = 0.90$. Observed inclusion concerns Δ^{cell}. The same collection supplies overlapping leave-one-out residual pools, so pooled inclusion is rank-constrained and is not independent validation of next-environment coverage. Candidate/family breakdowns and directional-error denominators are provided in the supplement.

Candidate	n	Nominal	Inclusion	Mean width	Median width	Commitment	$\mathrm{FA}_u/\mathrm{FA}_c$	$\mathrm{FF}_u/\mathrm{FF}_c$
TENT	2160	0.9000	0.9005	0.0426	0.0432	0.6787	0.0000/0.0000	0.0005/0.0028
EATA	2160	0.9000	0.9005	0.0340	0.0343	0.6356	0.0000/0.0000	0.0000/0.0000
SAR	2160	0.9000	0.9005	0.0260	0.0263	0.6694	0.0000/0.0000	0.0000/–

Tent regret is 0.0016, compared with 0.0080 for always-adapt and 0.1239 for always-freeze. EATA's values are 0.0013, 0.0033, and 0.1313. Their current actions are 1,107/359/694 and 1,241/132/787, respectively; both have zero observed false adaptations. SAR makes 1,446/0/714 decisions and has regret 0.0016, versus 0.0003 for always-adapt. A gate can avoid harmful updates yet lose opportunities when it abstains.

The six-family sensitivity analysis is less decisive than these point estimates. Tent's nominal family-bootstrap intervals favor KGA against each fixed policy, while EATA's adapt-side interval includes zero. Retrospective Holm adjustment over the six prospectively named candidate-by-baseline contrasts gives 0.09375 for both Tent comparisons. The replay, cluster unit, and test were chosen retrospectively and the families share one checkpoint. Full intervals and reference values appear in Table 13; this analysis is exploratory.

8.2 Observed Interval Inclusion Is a Separate Diagnostic

Commitment rate measures how often the interval excludes zero. Interval inclusion measures how often it contains the observed cell outcome. Table 6 reports these separately, together with full interval width. The calculation replays the current order-statistic rule from saved predictions and already-opened cell outcomes; it does not reuse historical stored actions or radii.

The nominal target is not a verified coverage guarantee for these dependent cells. Inclusion near 90% can follow from the rank procedure itself, even when transfer to a new family is poor. Family differences are useful diagnostics, but independent held-out-family, checkpoint, and natural-environment coverage remains a separate test.

All three candidates include 1,945 of 2,160 observed cell outcomes (90.05%). Zero observed false adaptations do not mean zero directional errors: Tent makes one false FREEZE among 359 FREEZE decisions. It occurs in the Gaussian-noise family, which has only two FREEZE decisions. EATA has no false FREEZE among 132 such decisions; SAR makes no FREEZE decisions, so its conditional false-freeze frequency is undefined. These denominators matter more than a pooled near-nominal inclusion fraction alone.

8.3 CCT-20: Conservative Retention, Not Selective Routing

Across 45 cells, KGA issued 0 ADAPT, 44 FREEZE, and 1 ABSTAIN decisions. It therefore served the frozen predictions in every cell. The candidate was helpful in 1 cell, tied in 0, and harmed in 44. The gate rejected most harmful updates but did not select the helpful one.

Table 7: CCT-20 safe-utility endpoint defined in Eq. (11). Contrasts are baseline regret minus KGA regret on the set-membership accuracy-proportion scale. These are nominal pointwise 95% two-way bootstrap intervals, not interval coverage for individual cell-benefit predictions. The stronger comparisons and location-level records remain in the supplement.

Comparator	Mean contrast	Nominal 95% CI	Required lower bound
Always adapt	0.1819	[0.0937, 0.2914]	$L > 0.0000$
Always freeze	0.0000	[0.0000, 0.0000]	$L > -0.0050$

KGA matched always-freeze exactly and avoided the measured degradation of always-adapt. The unchanged locked rule therefore returns pass on the stored cells, with generated flag yes. The recorded verdict remains SAFE UTILITY ONLY: a scoped safe-utility result, not a finite-sample population-safety guarantee. The stronger routing criterion failed because the gate never selected ADAPT and matched freeze. The zero false-adapt count (0/45) follows mechanically from zero adapt exposure.

All development cells were harmful for Tent, so the fit supplied no helpful example. Only two locations contributed fitting data, and calibration-to-target transfer is unverified. The complete target-location ledger and primary inference are retained in Appendix G.1; no new natural-shift success is inferred from this result.

8.4 Development Stops and Adverse Diagnostics

So2Sat-LCZ42 Zhu et al. (2020) tested 2 candidates on 9 development cities and 5 independent checkpoints. Tent had both helpful and harmful cities, but neither candidate passed all predeclared eligibility checks. The study stopped before gate calibration: no target inputs or labels were accessed, and there is no target natural-shift score. The development outcomes are now open, so retuning them cannot produce fresh confirmation.

The auxiliary studies also show limitations. ImageNet-R and PACS favor always-adapt; ImageNet-C is sensitive to the candidate and its operating point. Office-Home and RxRx1 mainly retain the frozen model, whereas the opened Camelyon17 OOD panel adapts throughout. These results are not hidden by the primary presentation. Appendix B reports the valid diagnostic rows on separate metric scales, together with the reasons why iWildCam, historical POEM/AETTA ports, and the focused Office-Home route cannot support current headline comparisons.

9 Limitations and Broader Impact

9.1 Assumptions, Not an Unconditional Safety Guarantee

The population frontier is exact within its declared rich binary class. A credible residual bound must come from domain knowledge, target labels, or a stated transfer assumption. A narrower class may permit stronger decisions, while an unsupported bound cannot justify a strict population claim. The useful object is the class-dependent feasible set, not the algebraic inequality alone.

For KGA, empirical interval coverage requires separate support at the intended decision unit. A larger radius reduces commitment but can increase regret when the frozen model is worse. A smaller radius increases exposure and makes calibration failures more consequential. Changing the adapter, checkpoint, feature schema, or update settings changes the prediction problem and requires recalibration or a justified transfer argument.

Repeated deployment needs a separate guarantee. A predeclared policy does not by itself supply exchangeability or time-uniform safety. Reusing calibration data, changing the frozen baseline after accepted updates, and selecting later candidates from earlier outcomes change the joint process. Adaptive conformal inference studies online coverage frequency with feedback Gibbs and Candes (2021); that is not time-uniform control of KGA's false commitments. Selection-conditional and group-conditional guarantees likewise require additional methods, as discussed in Section 2.

9.2 What Evidence Is Still Missing

The current controlled results share checkpoints and related corruption cells. The new interval-inclusion table is a retrospective diagnostic, not an independent calibration test. The natural results primarily evaluate retention; CCT-20 supplied no ADAPT exposure and So2Sat stopped in development. Empirical abstention may reflect weak prediction, limited data, wide intervals, or transfer failure. It does not establish population non-identifiability.

Four comparisons would strengthen the empirical claim: a current-policy replay against TTA/ALine, AETTA, POEM, and simple benefit-estimation baselines; a new natural-shift study with helpful and harmful updates and nonzero ADAPT/FREEZE exposure; interval inclusion and error evaluated on untouched groups; and inference across independently trained checkpoints and held-out environments. These are missing tests, not results obtained by rewriting or reclassifying the existing panel. The future protocol is specified in Appendix K.

9.3 Operational Scope and Potential Harm

A rejected update can prevent adaptation harm, but retaining a frozen model is not proof that the model is accurate or suitable for a high-stakes setting. Misinterpreting an abstention as a certificate would be particularly risky. Deployment should preserve the assessment reason and connect unresolved cases to a domain-appropriate review or evidence-acquisition process. The reported experiments do not validate clinical, safety-critical, or autonomous operation. No held-out physical camera deployment or common end-to-end latency result is claimed.

The maintained software distinguishes an unavailable assessment from a supported negative interval. Its tests and proof artifacts document a scoped validity, commitment-control, and rollback layer; they do not certify every data pipeline or production system. Reproduction instructions, formal scope, excluded historical extensions, and release checks are provided in the supplement.

10 Conclusion

K-Bound separates information that is absent from uncertainty that can be estimated. For a fixed model pair in the declared binary class, the feasible population benefit set determines when a strict direction is supported. The evidence-fibre audit result shows why the same unlabeled observations cannot remove the unresolved residual bound.

KGA is an empirical companion: it predicts a measured cell benefit and uses an interval to accept, reject, or defer an update. The controlled results show candidate-specific routing value. The CCT-20 result shows conservative retention, while the So2Sat development stop exposes a transfer limit. Together they motivate evidence-aware commitment, with the validity of each conclusion tied to its stated target and assumptions. Stronger natural-shift routing and independent coverage validation remain open empirical questions.

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A From Cell Coverage to Population Coverage

The interval proposition covers its declared target. The following conditional argument explains what would be needed to change a cell-level target into a population one.

Additional condition for population transfer. For cell outcomes using the same loss as the population risk, such as accuracy and 0/1 loss, suppose that, on the same probability space,

$$\mathbb{P}(|\widehat{\Delta} - \Delta^{\text{cell}}| \le \varepsilon) \ge 1 - \alpha_{\text{cell}},$$
$$\mathbb{P}(|\Delta^{\text{cell}} - \Delta| \le b) \ge 1 - \delta,$$

where $b \ge 0$ is available without the new evaluation labels, then

$$\mathbb{P}(|\widehat{\Delta} - \Delta| \le \varepsilon + b) \ge 1 - \alpha_{\text{cell}} - \delta.$$

The triangle inequality applies on the intersection of the two coverage events, and the union bound controls its complement; independence is not needed. A population certificate would use radius $\varepsilon + b$ and budget $\alpha_{\text{cell}} + \delta \le \alpha_{\text{target}}$. No such sampling-error bound is established for the reported panels, and their gates do not apply this enlarged radius. This conditional implication therefore adds no new population-safety claim to the empirical results.

For a nonlinear score such as macro-F1, an analogous argument must first define the population functional and a suitable sampling-error bound. Neither follows from the binary disagreement identity. Appendix L gives a counterexample with perfect cell-outcome coverage and negative population benefit.

B Supplementary Results and Non-Promoted Studies

This section preserves the evidence outside the two primary protocols. The presentation separates ordinary accuracy from balanced accuracy; no value is pooled across metrics, datasets, candidates, or protocols. The source-manifest identity is `266ce7cd87ae795e0742d43c40afefef82e82b555baa7dda8c6d270f73e2c03 1`. Historical, incomplete, and invalid studies keep their original scope instead of being silently removed.

B.1 Accuracy-Based Diagnostic Replays

Table 8 collects the remaining ordinary-accuracy replays; the dataset-specific interpretations follow below.

Table 8: Supplementary accuracy-regret rows. Lower regret is better. Commitment rate counts ADAPT or FREEZE and is not interval coverage. These are protocol-specific point summaries: Office-Home's primary and stream-seed rows are different from its invalidated five-checkpoint audit; ImageNet-C is configuration-sensitive; PACS has aggregate-only replay. For PACS, $n = 12$ counts domain–seed inference units, whereas action frequencies use 216 repeated decisions.

Protocol	n	KGA	Adapt	Freeze	FA_{u}	Commitment rate
Office-Home primary	35	0.0158	0.0468	0.0158	0.0000	0.3143
Office-Home stream seeds	54	0.0215	0.0458	0.0217	0.0000	0.2778
ImageNet-C Tent	135	0.0145	0.0191	0.0145	0.0000	0.0000
ImageNet-C EATA	135	0.0007	0.0001	0.0342	0.0074	0.8963
ImageNet-C SAR	135	0.0289	0.0529	0.0319	0.0074	0.2074
PACS (aggregate)	12	0.0431	0.0176	0.0446	0.0093	0.8148
CIFAR-10.1	48	0.0017	0.0190	0.0017	0.0000	0.4583

B.2 Balanced-Accuracy Diagnostic Replays

Table 9 reports the balanced-accuracy replays on their own metric scale, with each recorded candidate and evaluation unit kept distinct.

Table 9: Supplementary balanced-accuracy-regret rows, kept separate from ordinary accuracy. The archived runner contracts name balanced accuracy for Camelyon17, ImageNet-R, and RxRx1; ordinary accuracy can coincide only under the recorded class-balancing conditions. ImageNet-R aggregates candidate backbones rather than one deployed gate. Camelyon17 OOD was already opened; its B–v2 rows are separate within-seed diagnostics. RxRx1's displayed row is model seed 0.

Protocol	n	KGA	Adapt	Freeze	$\mathrm{FA_u}$	Commitment rate
ImageNet-R backbones	480	0.0150	0.0064	0.0325	0.0000	0.4042
Camelyon17 OOD	18	0.0000	0.0000	0.1381	0.0000	1.0000
Camelyon17 B–v2 Tent	108	0.0296	0.0097	0.0820	0.0093	0.3704
Camelyon17 B–v2 EATA	108	0.0083	0.0040	0.0911	0.0000	0.5648
Camelyon17 B–v2 SAR	108	0.0006	0.0016	0.1001	0.0000	0.6574
RxRx1 model seed 0	60	0.0000	0.2531	0.0000	0.0000	1.0000

Table 10: Office-Home protocols are different experiments, not conflicting versions of one row. Entries are measured accuracy regret to the per-condition oracle; lower is better. The first row is the canonical single-candidate KGA replay with fixed SAR. The focused route is invalid, so only the post-hoc fixed-SAR opportunity regret (.0027) and freeze regret (.0230) are retained; its actions have no routing or certificate interpretation. A/F/U are adapt/freeze/abstain actions.

Protocol	n	Router / Adapt / Freeze	A/F/U
Archived stream-seed replay	54	.0215 / .0458 / .0217	1/14/39
Five-checkpoint candidate audit	15	invalid / .0027 / .0230	n/a

B.3 Large-Scale Corruption: ImageNet-C

ImageNet-C is candidate-dependent. SAR has KGA/adapt/freeze regrets 0.0289/0.0529/0.0319, a pooled point advantage relative to each fixed policy, but with one false adaptation among 135 cells and without seed-robust inferential support. Tent ties the always-freeze point estimate and makes no strict decisions in the reconciled row. EATA nearly tracks always-adapt but has a larger KGA regret (0.0007 versus 0.0001). These results show why ImageNet-C must be interpreted at the candidate-policy level rather than through one dataset-wide success label. This SAR result uses a learning rate 16× higher than SAR's published setting (4×10^{-3} versus 2.5×10^{-4}), with the final ResNet stage left adaptable—an aggressive configuration at which SAR is expected to be fragile. No control at SAR's official operating point has been run; this is a claim about this configuration, not about SAR as published.

B.4 Office-Home

The primary Office-Home row Venkateswara et al. (2017) is conservative. KGA regret is 0.0158, tying always-freeze (0.0158) and below always-adapt (0.0468), with 0 adapt decisions. This is a descriptive point-estimate match that evaluates conservative retention rather than selective routing. The archived protocol does not supply a predeclared uniform stability guarantee for the full-fit versus out-of-fold benefit predictor.

The 54-cell Office-Home stream-seed replication in Table 8 has KGA regret 0.0215 versus 0.0458 and 0.0217, a small point advantage relative to each fixed policy. The three run seeds do not record independent checkpoint identities, and the seed-level interval includes zero, so the comparison remains descriptive.

The post-hoc five-checkpoint Office-Home study is a candidate opportunity audit, not a valid KGA result. It uses five independently trained source checkpoints, the three target-test domains, one fixed IID/tiny condition, and Tent, EATA, and SAR. Checkpoint identity passed, but routing and serialization did not. The identifying agreement relation is binary-only and does not apply to the 65-class task; the old spectral estimate was sign-indeterminate and unbounded. Thirteen of 15 anchors were negative, 9 of 60 stored $\widehat{b}$ values fell outside $[0, 1]$, and EATA and SAR produced exactly the same predictions in 10 of 15 cells. Each checkpoint has only three LOO cells, whereas a 90% empirical order-statistic single-candidate radius requires at least 10 total cells. All five result files fail strict JSON and contain 57 literal `Infinity` values in total.

The archived exploratory implementation printed ABSTAIN in all 15 cells, but those outputs have no safety, routing, or certificate meaning. The valid descriptive quantities are checkpoint accuracies and candidate opportunity: freeze is 0.5711, SAR is 0.5914, their paired difference is 0.0203 (nominal post-hoc 95% t-interval $[0.0123, 0.0283]$), and the per-cell oracle is 0.5941. Oracle minus SAR is 0.0027 (nominal post-hoc 95% interval $[-0.0024, 0.0077]$). The best adapted candidate improves over freeze in 13 cells, ties in one, and harms in one. Because the target was already opened and SAR was chosen after comparing the audit means, these statistics are post-hoc opportunity evidence only.

B.5 Camelyon17

The archived Camelyon17 OOD evaluation Bandi et al. (2019) yields zero KGA and always-adapt regret against 0.1381 for freeze. All 18 cells are helpful, so this already-opened panel is a descriptive always-adapt one-sided diagnostic rather than prospective evidence of selective routing. Within-seed candidate diagnostics remain separate from this OOD endpoint. In the B–v2 SAR diagnostic, KGA regret is 0.0006, compared with 0.0016 for always-adapt and 0.1001 for always-freeze. All three run-seed point estimates favor KGA relative to each fixed policy, but the archived paired interval against always-adapt includes zero and the Holm-adjusted comparison does not pass. The run seeds also do not record independent checkpoint identities, and the target was already open. We therefore retain this as within-seed stress evidence rather than untouched natural-domain transfer.

B.6 RxRx1

The displayed RxRx1 row Sypetkowski et al. (2023) is model seed 0: KGA and freeze regret are zero against 0.2531 for always-adapt, with no adapt decisions. A separate audit across three independently trained model checkpoints repeats the same all-freeze tie. It strengthens model-variation reproducibility but adds no adapt exposure. RxRx1 therefore evaluates reproducible conservative retention rather than selective routing.

B.7 iWildCam

iWildCam Koh et al. (2021) is not assigned a canonical numerical row. The archived scorer used sklearn macro-F1, which averages prediction-only classes, rather than the official WILDS metric, which averages labels present in the target truth. The archived runtime, sample identities, and target population were not sealed under the current contract, so its diagnostic scores and routing actions remain audit-only. A pinned rerun with the official metric, sealed sample identifiers, and a population hash is required before promotion. The older binary dominance flag and the later replay under the same invalid metric contract remain hash-locked only as superseded audit history.

B.8 Boundary and Transfer Diagnostics

B.8.1 PACS

PACS Li et al. (2017) KGA regret is 0.0431, compared with 0.0176 for always-adapt. Its archived aggregate lacks the residual detail needed for full gate replay, so it remains an aggregate performance diagnostic with incomplete certificate replay.

B.8.2 ImageNet-R Backbone Breakdown

ImageNet-R Hendrycks et al. (2021) KGA regret is 0.0150, compared with 0.0064 for always-adapt, across the ten-backbone architecture panel. No backbone has lower KGA point regret than each fixed comparator. KGA regret is higher than always-adapt regret on 8 of 10 backbones; four of those eight have a 0% harmful base rate, so routing has no harm to avoid. On `efficientnet_b0` (100% harmful) and `swin_t` (60.4% harmful), KGA improves over always-adapt but exactly matches always-freeze and makes no adapt decisions. The pattern is consistent with the regime map: routing can add value only where candidate harm is present, while these two rows evaluate conservative retention rather than selective routing.

B.8.3 CIFAR-10.1

CIFAR-10.1 Recht et al. (2018) matches always-freeze at 0.0017 and is retained as a low-margin cross-seed transfer diagnostic. This pattern alone does not prove structural non-identifiability; weak evidence, low margin, estimator inadequacy, or calibration transfer can produce the same behavior.

B.9 Historical Protocol-Matched POEM and AETTA Ports

The repository retains a head-to-head artifact on the locked CIFAR-10-C stress stream, but its KGA actions came from an earlier policy and are not synchronized with the current empirical order-statistic actions. Its numerical ordering is therefore audit history, not release-eligible comparative evidence, and we do not report a superiority estimate from it. The neighboring methods are protocol-matched ports, not untouched official implementations. A defensible comparison requires all methods to be replayed on identical current-policy records with the multiplicity family fixed in advance.

These controlled results use empirical calibration. Leaving one cell out prevents that cell from directly fitting its own prediction or radius. It does not establish exact split-conformal coverage when corruption cells are dependent. Thus zero observed false adaptations describes this protocol; it does not guarantee the same result for every future corruption family.

B.10 So2Sat Development and Action-Unit Audit

The locked So2Sat-LCZ42 v1 study evaluated 2 candidates, Tent and SAR, on 9 development cities and 5 independent source checkpoints, giving 45 city–checkpoint cells per candidate. This was a candidate-feasibility stage, not target evaluation. Tent had 4 helpful and 4 harmful cities and a cell-oracle opportunity of 0.8848 percentage points. Its leave-one-city-out sign agreement was 51.11%, and its routed score was 0.1544 percentage points below always-adapt, its better fixed policy. SAR had 0 qualifying helpful cities, 6 harmful cities, and an oracle opportunity of 0.5640 percentage points. It selected ADAPT in only 5 of 45 cells, and its routed score was 0.0842 percentage points below always-freeze. Tent passed 7 of 9 feasibility checks and SAR passed 6 of 9, so neither candidate was eligible.

The locked status records no feasible candidate and a stop before gate calibration. No held-out gate-calibration rows or target inputs were accessed; the recorded counts were 0 gate-calibration rows, 0 target inputs, 0 target pixels, and 0 target labels. The study therefore documents failure to meet the v1 cross-city development-transfer feasibility criteria while also showing that a cell-oracle routing opportunity existed for Tent. It contains no target natural-shift score. All 90 development outcomes are now open, so retuning on them cannot produce fresh confirmatory evidence. In addition, v1 declares one target action per city while the target runner creates city–checkpoint actions. This mismatch did not affect the stop, because no target action was created, but target execution remains blocked until a new versioned protocol and implementation use the same action unit.

B.11 Withdrawn Development Proxy for the Population Budget

The frontier is exact relative to $\mathcal{C}_\beta$, but an unlabeled deployment batch cannot verify $|\gamma| \le \beta$. A credible budget must come from domain knowledge, a declared coupling, or historical labeled deployments under a transfer assumption. Setting $\beta = 0$ assumes a zero target disagreement-conditional residual; it is not a safe default.

We tested a source-development benefit-scale proxy as a possible operational surrogate for β; it does not estimate the theorem's calibration-residual bound. On CIFAR-10-C, the proxy is 1.4–50$\times$ too small relative to evaluation-scale discrepancies: 24–73% of cells violate the putative budget, and commit error is 0.4–16.6%. On ImageNet-C, a larger proxy often removes most decisions; 5 of 10 configurations return zero commitments on all 405 cells. We therefore withdraw this surrogate route. The population β remains externally supplied from domain knowledge or a stated assumption.

Table 11: Claim-to-support ledger for the compact submission. Support pairs each evidence reference with the conditions required for its stated scope.

Claim and type	Support and requirement	Boundary
Interior evidence ambiguity (theorem)	Thm. 1; $\beta > 0$, $\lvert M\rvert < \beta$, and full fibre richness	binary, 0/1 loss
Closed-band abstention (proposition)	Prop. 1; strict directional semantics and a rich class	zero benefit blocks a strict claim
Strict-commitment frontier (theorem)	Thm. 2; fixed augmented evidence and declared $\mathcal{C}_\beta$	β is externally supplied
Marginal false-adapt control (operational proposition)	Prop. 2 under Eq. (9) for the measured cell target	neither population-risk nor conditional FA_c control follows
CIFAR-10-C Tent regret (empirical)	Tables 5 and 13; frozen Tent controlled-grid protocol	lower empirical order-statistic point regret; nominal family-bootstrap intervals favor KGA; retrospective Holm is nonconfirmatory
CIFAR-10-C EATA regret (empirical)	Tables 5 and 13; frozen EATA controlled-grid protocol	adapt-side family interval includes zero; retrospective Holm is nonconfirmatory
Natural-shift conservative rows (empirical)	Tables 8, 9, and 7; each split and candidate remain fixed	often zero ADAPT exposure
So2Sat development stop (diagnostic)	Sec. 8.4; locked v1 feasibility rules and unopened target boundary	no eligible candidate, gate calibration, or target score
ImageNet-C candidate scope (diagnostic)	Table 8; candidate rows interpreted separately	candidate-specific evidence
PACS/ImageNet-R/CIFAR-10.1 scope (diagnostic)	Tables 8 and 9; canonical replay scope retained	observed behavior does not prove unknowability
Deployment generalization (future evidence)	Sec. K; diverse unopened natural targets	outside current evidence scope
Real-camera validation (pending)	locked sessions and fresh held-out recordings	protocol infrastructure only; no result in this paper

The theory-to-algorithm link needs a sampling argument. The population layer targets Δ; the reported empirical calibration targets Δ^{cell}, as Section 6.1 makes explicit. Coverage for the latter does not cover the former without an additional evaluation-error argument such as that in Appendix A. Neither layer establishes that Z is risk-aligned, that h_θ transfers, or that residual coverage holds in a new domain. KGA applies the three-way decision through an interval; it does not learn these validity assumptions.

C Compact Assumption and Claim Ledger

Table 11 separates formal claims from empirical findings and records the conditions and scope limits for each.

D Calibration and Evaluation Contracts

Controlled-grid contract. The fitting unit is the condition cell. An out-of-fold benefit estimate is produced without fitting on the scored cell, and the residual radius for that cell excludes its residual. This is leave-one-condition-out cross-fitted empirical residual calibration. Related cells may still be dependent through corruption family, checkpoint, or seed. Exact split-conformal wording is not used for this branch.

Natural-shift contract. Development conditions fit the benefit model; a disjoint calibration partition supplies residuals; choices are frozen; and the target partition is scored once where the archived protocol records all three stages. When the saved evidence does not support that complete chain, the row is marked descriptive or diagnostic. Replaying a table does not strengthen its original design.

The candidate TTA step itself is transductive. Episodic candidates update on each unlabeled evaluation batch before predicting that batch. Online candidates update on a separate auxiliary stream, but prediction still

Table 12: Action exposure in replayable canonical rows. Counts are not independent experiments: they are evaluation units within the protocol shown. A zero false-adapt count is informative only when adaptation was attempted and the residual pool did not make the inequality automatic. Harm and false adaptation are defined by measured cell outcomes, not population risks. Counts were regenerated from source manifest SHA-256 `266ce7cd87ae795e0742d43c40afefef82e82b555baa7dda8c6d270f73e2c031`.

Row	n	A	F	U	false A	FA_u	repetition / scope
Office–Home primary	35	0	11	24	0	.0000	2 test-stream seeds; ties freeze
Office–Home stream-seed repl.	54	1	14	39	0	.0000	3 test-stream seeds; point-only edge
ImageNet–C SAR	135	13	15	107	1	.0074	5 seeds × 27 cells
ImageNet–C Tent	135	0	0	135	0	.0000	5 seeds × 27 cells
ImageNet–C EATA	135	120	1	14	1	.0074	5 seeds × 27 cells
ImageNet–R	480	165	29	286	0	.0000	4 seeds × 10 backbones × 12 cells
CIFAR–10–C Tent	2160	1107	359	694	0	.0000	5 seeds × 432 cells
CIFAR–10–C EATA	2160	1241	132	787	0	.0000	5 seeds × 432 cells
CIFAR–10–C SAR	2160	1446	0	714	0	.0000	5 seeds × 432 cells
Camelyon17 OOD	18	18	0	0	0	.0000	archived opened helpful-only OOD subset
Camelyon17 B–v2 Tent	108	40	0	68	1	.0093	3 within-seed grids; diagnostic
Camelyon17 B–v2 EATA	108	61	0	47	0	.0000	3 within-seed grids; diagnostic
Camelyon17 B–v2 SAR	108	71	0	37	0	.0000	3 within-seed grids; diagnostic
RxRx1 J	60	0	60	0	0	.0000	displayed model seed 0; separate 3-checkpoint audit also all-freeze
CIFAR–10.1 K	48	0	22	26	0	.0000	locked cross-seed replay

uses evaluation-batch BatchNorm statistics. Thus “disjoint” above refers to development, residual-calibration, and target partitions (and to the recorded auxiliary stream versus evaluation sample identities); it does not mean that candidate inference is inductive. No target labels enter adaptation, prediction, evidence, or the action.

Action contract. Every action is computed from Z, $\widehat{\Delta}$, and ε without target labels. Offline labels determine Δ^{cell}, the measured oracle action, empirical regret, and false-commit indicators only after the action is fixed. Existing artifact fields named `delta` store this cell outcome; their schema and numerical values are unchanged. Abstention resolves to the frozen prediction for measured task-performance accounting, while its log retains the distinction from a certified freeze.

Configuration contract. The checkpoint, candidate adapter, optimizer, adapted parameter subset, update count, evidence schema, feature normalization, benefit estimator, calibration split, α, and software version are part of one calibration identity. Changing any of them requires either a declared transfer argument or recalibration. The fail-closed default on identity mismatch is non-commitment.

E Decision Exposure and Repetition Audit

Point regret alone can conceal whether a conservative match exercised the ADAPT branch. Table 12 therefore reconstructs action counts from the same canonical panel used in Tables 5, 8, and 9. A/F/U denote adapt, freeze, and abstain. PACS is omitted from the action table because the released aggregate cross-checks regret but does not contain the per-cell benefit estimates and residuals required for a complete gate replay. iWildCam is omitted because its archived actions were scored under the wrong macro-F1 contract.

The largest adapt exposure occurs on the controlled CIFAR-10-C rows. This is why those rows are the main empirical test of the adapt-side certificate. Primary Office-Home and RxRx1 have zero adapt exposure, while the 54-cell Office-Home stream-seed replication has one adapt decision. Their zero false-adapt values barely test whether the gate can distinguish a harmful adaptation from a helpful one; they show that the controller declines the candidate under those evidence/calibration contracts. The invalidated five-checkpoint Office-Home route and the withheld iWildCam actions are not counted. Camelyon17 OOD has the opposite limitation: every archived condition is helpful and every action adapts, so there is no harmful opportunity to reject. ImageNet-C EATA has substantial adapt exposure and one false adaptation, while Tent abstains everywhere. Those three patterns are operationally different even when their regrets are small.

Table 13: Retrospective current-policy sensitivity over six observed CIFAR-10-C corruption families. Each cell reports baseline regret minus KGA regret and a nominal, unadjusted 95% family-bootstrap interval; positive values favor KGA. Holm values adjust retrospective sign-flip reference values over six prospectively named candidate-by-baseline contrasts. The replay, corruption-family unit, sign-flip test, and Holm adjustment were not preregistered. The within-Tent two-contrast Holm value is 0.03125 but is post hoc. This table is conditional on one archived checkpoint and does not provide verified finite-sample coverage, simultaneous intervals, independent-checkpoint inference, prospective confirmation, natural-shift evidence, or official POEM/AETTA superiority.

Candidate	Adapt gap [95% CI]	Freeze gap [95% CI]	Six-way Holm p (A/F)
Tent	0.00633 [0.00254, 0.00954]	0.12229 [0.06331, 0.17772]	0.09375/0.09375
EATA	0.00195 [-0.00052, 0.00438]	0.13001 [0.06977, 0.18754]	0.25000/0.09375
SAR	-0.00122 [-0.00267, -0.00007]	0.13891 [0.08031, 0.19686]	0.90625/0.09375

The repetition column also prevents a common inflation of evidence. Five seeds of one checkpoint and grid protocol are not five independently designed studies. Ten ImageNet-R backbones are a panel of candidate policies, not ten target domains for one frozen gate. Three Camelyon17 within-seed grids permit a stability diagnostic but do not create an untouched hospital-domain replication. RxRx1 is stronger on model variation because the source checkpoint is independently trained three times, but its all-freeze behavior still provides no adapt exposure.

F Inference Discipline

F.1 Point Estimates, Paired Gaps, and Clusters

For a baseline $b \in \{\text{always-adapt}, \text{always-freeze}\}$, the paired routing gap at condition j is

$$G_{bj} = \text{regret}_{b,j} - \text{regret}_{\text{KGA},j} \,.$$

A positive mean gap means lower regret for KGA. All paired routing-gap intervals reported in this paper use this baseline-minus-KGA sign convention. The point-estimate two-comparator criterion requires both means to be positive. Interval support additionally requires the lower confidence limit for both paired means to exceed zero at the declared resampling unit. These definitions are intentionally separate: Table 8 can establish a point ordering, but not an inference claim without the associated unit, interval artifact, and justified coverage assumptions. A positive computed lower endpoint is numerical interval support under the resampling model, not verified coverage. Multiplicity adjustments do not repair invalid marginal intervals or tests.

Condition-level resampling treats cells as exchangeable and can understate uncertainty when severities and compositions share a corruption family. Corruption-family resampling preserves observed within-family dependence, but few clusters limit reliability; conservativeness is not guaranteed. Seed resampling measures variation across repeated training or test streams only when the seeds represent the intended deployment randomness. The compact paper therefore does not use one generic "bootstrap CI" label. An inference claim must state whether the unit is cell, corruption–severity pair, corruption family, seed, backbone, domain, or physical session.

F.2 Current-Policy Corruption-Family Sensitivity

The six candidate-by-baseline contrasts were named before this analysis; the replay policy, corruption-family unit, sign-flip reference test, and Holm adjustment were selected retrospectively. The current-policy artifact averages run seeds and condition cells within each of the six families. Table 13 retains every contrast, including EATA's adapt-side interval crossing zero and SAR's disadvantage against always-adapt. The within-Tent two-contrast Holm calculation is post hoc and is not the declared six-contrast family.

Table 14: Current-policy interval and decision diagnostics by corruption family. Width is full interval width, 2ε. Inclusion is measured on already-open, dependent cross-fitted cells. Conditional errors use the number of the corresponding action as denominator and are undefined when that action is absent. These diagnostics are not held-out group-coverage validation.

Candidate	Family	n	Inclusion	Mean width	Commitment	False A/A; FA_c	False F/F; FF_c
TENT	contrast	360	0.7972	0.0425	0.7028	0/180; 0.0000	0/73; 0.0000
TENT	defocus_blur	360	0.9417	0.0426	0.7222	0/180; 0.0000	0/80; 0.0000
TENT	fog	360	0.9333	0.0426	0.7139	0/180; 0.0000	0/77; 0.0000
TENT	gaussian_noise	360	0.8556	0.0425	0.8611	0/308; 0.0000	1/2; 0.5000
TENT	jpeg_compression	360	0.9639	0.0426	0.3944	0/79; 0.0000	0/63; 0.0000
TENT	pixelate	360	0.9111	0.0426	0.6778	0/180; 0.0000	0/64; 0.0000
EATA	contrast	360	0.8333	0.0340	0.5778	0/180; 0.0000	0/28; 0.0000
EATA	defocus_blur	360	0.9306	0.0341	0.6028	0/180; 0.0000	0/37; 0.0000
EATA	fog	360	0.9083	0.0340	0.5972	0/180; 0.0000	0/35; 0.0000
EATA	gaussian_noise	360	0.8472	0.0340	0.9833	0/354; 0.0000	0/0; –
EATA	jpeg_compression	360	0.9667	0.0341	0.4944	0/167; 0.0000	0/11; 0.0000
EATA	pixelate	360	0.9167	0.0340	0.5583	0/180; 0.0000	0/21; 0.0000
SAR	contrast	360	0.8778	0.0260	0.5000	0/180; 0.0000	0/0; –
SAR	defocus_blur	360	0.9167	0.0260	0.5000	0/180; 0.0000	0/0; –
SAR	fog	360	0.9028	0.0260	0.5000	0/180; 0.0000	0/0; –
SAR	gaussian_noise	360	0.8306	0.0260	1.0000	0/360; 0.0000	0/0; –
SAR	jpeg_compression	360	0.9583	0.0261	0.9639	0/347; 0.0000	0/0; –
SAR	pixelate	360	0.9167	0.0260	0.5528	0/199; 0.0000	0/0; –

F.3 Retrospective Interval Inclusion by Family

The following audit replays the current policy from stored, cross-fitted predictions and measured cell outcomes. It uses each candidate's original residual rule and excludes the scored residual from its own radius. No fitting, tuning, new target access, or training is performed. The input, policy, code, and output digests are recorded in the generated diagnostic JSON.

The pooled inclusion fraction is constrained by the overlapping leave-one-cell-out order statistics. It is not an independent estimate of coverage on new environments. Family rows reveal the distribution of those same misses, widths, and decisions; they do not supply group-conditional guarantees. A nominal 90% target is a calibration setting, not a promise validated by this replay.

F.4 False-Adapt Accounting

The event in Eq. (4) is unconditional and concerns measured cell benefit. Its reported frequency divides ADAPT decisions with $\Delta_j^{\text{cell}} \leq 0$ by all evaluated units, including freezes and abstentions. This need not estimate the population-harm event $\{g = \textsc{adapt}, \Delta \leq 0\}$. Conditional false-adapt divides by adapt decisions and is undefined when no adaptation occurs. A zero observed count should be accompanied by the number of adapt decisions; otherwise a controller that never adapts appears perfect. Table 12 exposes that denominator.

If the scored residual is included in the same finite residual vector whose order statistic decides the action, the number of residuals above the order statistic is algebraically constrained. Such coverage is not an independent empirical validation. Excluding the scored residual removes direct self-inclusion, but pooled LOO inclusion remains rank-constrained. It is not independent validation and does not supply the exchangeability or stability needed for a finite-sample theorem. Clean split calibration and LOO empirical calibration therefore remain different protocol labels.

F.5 Multiplicity and Search History

An earlier repository census counted many historical two-comparator flags, but it did not freeze the file list and commit needed to reproduce a current total. We therefore do not promote that count or its positive rate as release evidence. The panel in this paper is a post-hoc reconciled audit panel assembled from individually locked and historical protocols; it is not a globally preregistered confirmatory family. A result selected after a search cannot become confirmatory by correcting only the final positive rows. Future confirmatory

Table 15: What is present in the compact release and what remains unavailable. "Replay" means the stored quantities suffice to recompute the displayed gate output under the released scorer.

Track	Released status	Present evidence	Remaining limitation
CIFAR-10-C	replayable	per-condition Δ^{cell}, $\widehat{\Delta}$, radius, action; 5 seeds; 3 candidates	related grid cells; no untouched corruption-family test in canonical table
ImageNet-C	replayable	per-condition records for 5 seeds and Tent/EATA/SAR	27-cell controlled configuration; candidate-specific, not one policy
Office-Home	replayable	canonical primary and 54-cell stream-seed records; later five-checkpoint audit reported separately	full-fit-versus-OOF transfer-stability premise not predeclared
iWildCam	withheld	archived opened test-split records; non-promoted official-metric diagnostic	wrong archived metric; population was not sealed and overlap is unauditable
Camelyon17	replayable diagnostic	archived opened OOD row plus separate B–v2 diagnostics	OOD row is one-sided and non-prospective; B–v2 is not untouched-domain evidence
RxRx1	replayable	displayed model-seed-0 row plus three model-checkpoint records and source hashes	all three checkpoints freeze throughout; no helpful test conditions
CCT-20	sealed complete	5 independent checkpoints, 9 target locations, actions, predictions, score, and inference hashes	only two fit locations; no ADAPT exposure; trans-target extrapolation
So2Sat-LCZ42	sealed development stop	2 candidate bundles, 9 development cities, 5 checkpoints, selection record, and 0 target inputs	no feasible candidate; all 90 gate-fit outcomes now open; no gate calibration or target result
PACS	partial	three-seed aggregate and source cross-checks	missing per-cell $\widehat{\Delta}$ and residuals prevent full gate replay
ImageNet-R	replayable	4 seeds, 10 candidate backbones, 12 conditions each	architecture aggregate is diagnostic, not one deployable router
CIFAR-10.1	replayable	locked cross-seed condition records	low-margin conservative match; selective routing not exercised
Physical camera	protocol only	dashboard, logging schema, planned session split	no fresh held-out physical recordings or result table

families must name every candidate–track test before any test value is computed, and must retain failures and incomplete runs.

Search history. The repository contains many exploratory configurations. One superseded review census reported 1,387 determinations and 326 positives (23.5%); a broader archived scan reported 1,427 determinations and 345 positives. Because neither census froze one reproducible family definition, file list, and commit, neither total is promoted as the release multiplicity family. We instead report a current post-hoc reconciled panel and do not relabel its favorable rows as confirmatory discoveries. Future claims should declare the candidate, calibration unit, target unit, seed structure, inference unit, and two-comparator success criterion before test labels are opened.

G Track-Level Provenance and Remaining Evidence

Separate archival records preserve two additional scope checks outside the canonical panel. FMoW Protocol L reached held-out scoring and matched always-freeze, while always-adapt had lower regret; its status remains not-cleared. PovertyMap stopped at its preregistered development screen because the harm-detection AUC was 0.637, below the 0.65 threshold, so held-out validation and test scoring did not run. These records remain in the claim inventory and are not pooled with the canonical rows.

The provenance table is part of the result, not administrative metadata. A reviewer can trace a canonical row to compact records and their original archive hashes. If the original large artifact is not shipped, the manifest still records its byte count and digest. This guards against silent replacement of a result while allowing the release to omit raw datasets and oversized transient checkpoints.

Three upgrades would materially strengthen the empirical case. First, hold out entire shift families rather than individual controlled cells and fit all hyperparameters before opening the family. Second, add independent model and test-session seeds to the natural tracks that currently have one target replay. Third, run the physical protocol with days, devices, and objects assigned at the session level, then score every policy on the

Table 16: Locked CCT-20 primary contrasts. The estimand is baseline regret minus KGA regret, so positive values mean lower regret for KGA. Intervals use a paired two-way product bootstrap that resamples checkpoint rows and location columns independently. Two nominal 97.5% per-comparison intervals target a nominal 95% Bonferroni family level; actual family coverage requires valid marginal coverage. Reference p-values enumerate one-sided sign flips of the nine location means under Eq. (12). Holm adjusts only these reference values; the threshold flags reproduce the locked calculation, not a verified population guarantee.

Comparator	Baseline regret − KGA regret	Nominal 97.5% CI	Sign-flip p	Holm p	Holm flag (.05)
Always adapt	0.18190227	[0.08501071, 0.31066374]	0.00195312	0.00390625	yes
Always freeze	0	[0, 0]	1	1	no

same locked streams. These upgrades address dependence and transfer; simply adding more rows from the same grid does not.

G.1 CCT-20 Inference, Location, and Release Ledger

Provenance ledger. Cell-level outcomes remained unopened before target execution. Aggregate target label metadata had already been inspected during candidate ranking, so this is not a label-unopened study. Item-level target labels were deserialized once, after the one-shot scoring marker was spent. The release chain also records the numeric-v2 repair in a sealed addendum created before outcome access. That repair replaced an unstable divergence calculation with a stable Gaussian KL computation; it did not change the model, data split, or scoring rule. We treat it as protocol provenance, not result-driven tuning. All primary outcome values and action/effect counts come from the sealed, content-addressed, receipt-linked CCT-20 release manifest and its generated macros (canonicalized inference-document SHA-256 `5d1ba74d9a8b16ef24746ca3d95969e635d34fd4112806f6bf91b4b45a6cf907`).

Finite-sample sign-flip validity requires the null joint law of the nine location gaps to satisfy Eq. (12). Independent zero-symmetric location gaps conditional on the evaluated checkpoints would suffice, but that assumption is unverified here. Exchangeability alone does not suffice. The sign-flip analysis conditions on the evaluated checkpoints, whereas the two-way bootstrap also resamples checkpoint levels. These are different uncertainty models, and neither establishes distribution-free safety for unseen locations.

Primary result. KGA exactly matched always-freeze. Its contrast against always-freeze was 0 with nominal Bonferroni-adjusted 97.5% bootstrap interval [0, 0], enumerated location sign-flip $p = 1$, and Holm $p = 1$. Against always-adapt, the contrast was 0.18190227 with interval [0.08501071, 0.31066374], sign-flip reference $p = 0.00195312$, and Holm $p = 0.00390625$. Measured cell benefit is adapted accuracy minus frozen accuracy: 1 cell was helpful, 0 were exactly zero, and 44 were harmful. These counts explain why freezing was useful here. They do not show that the gate learned when to adapt.

Limits and next step. All 55 development cells showed lower performance after Tent than under the frozen model, so the gate had no helpful ADAPT example. The ridge fit used 10 checkpoint–location rows from only two camera units for 11 recorded features. Calibration used 45 rows collapsed to nine cis-test camera locations, while the locked targets came from trans-test locations. Coverage therefore needs an unverified cross-split exchangeability assumption. Target Mahalanobis diagnostics ranged from 7.12 to 485.67 (median 28.73), showing strong extrapolation; this diagnostic never changed an action and is not a formal coverage guarantee. One natural target study also cannot establish transfer of selective routing across natural shifts. A follow-on locked study on So2Sat-LCZ42 retained an outcome-blind boundary but stopped at development feasibility, as described next. A new versioned study still needs diverse development units, useful exposure to both ADAPT and FREEZE, and untouched confirmation environments. Cell-level 16-indicator macro-F1 is archived as descriptive secondary evidence; no post-hoc aggregate or inference claim is made.

Table 17 reports all nine location clusters. Evaluation n is the number of evaluation images per checkpoint at that location. Each A/F/U triplet and each helpful/zero/harmful triplet therefore has denominator five, one cell for each checkpoint. These labels describe measured cell outcomes. Accuracy differences use one sign

Table 17: Complete locked CCT-20 location ledger. A/F/U denotes ADAPT/FREEZE/ABSTAIN, and +/0/− counts positive/zero/negative measured-benefit cells. Difference columns are equal-weight means over five checkpoint cells and are rounded to four decimals for display; the sealed manifest retains full precision. Cells are repeated-model observations, and the inferential cluster unit is the camera location.

Location	n/checkpoint	A/F/U	Benefit +/0/−	Adapt − freeze	KGA − adapt	KGA − freeze
0	1302	0/5/0	0/0/5	-0.5418	0.5418	0.0000
7	1283	0/5/0	1/0/4	-0.0293	0.0293	0.0000
28	911	0/5/0	0/0/5	-0.0562	0.0562	0.0000
40	800	0/5/0	0/0/5	-0.1232	0.1232	0.0000
46	4432	0/4/1	0/0/5	-0.0723	0.0723	0.0000
78	1633	0/5/0	0/0/5	-0.0839	0.0839	0.0000
100	3441	0/5/0	0/0/5	-0.2035	0.2035	0.0000
105	1540	0/5/0	0/0/5	-0.2892	0.2892	0.0000
130	983	0/5/0	0/0/5	-0.2376	0.2376	0.0000

throughout: adapted minus frozen for adaptation benefit, and KGA minus the named fixed policy for the two policy columns.

The release ledger contains all 45 cells: 0 ADAPT, 44 FREEZE, and 1 ABSTAIN actions; 1 helpful, 0 zero, and 44 harmful adaptation effects. The release verdict remains SAFE UTILITY ONLY. The location table is descriptive; sign-flip enumeration uses the nine location means under the reference model in Eq. (12), and no post-hoc aggregate is claimed for the secondary macro-F1 outcome.

H Extended KGA Implementation Contract

H.1 State Isolation

The frozen and candidate predictors must not share mutable normalization buffers, optimizer state, or parameter storage during preview. The candidate is created from a checkpoint, adapted on the unlabeled window, and evaluated without overwriting the active state. Only an adapt decision may replace the active checkpoint. Freeze and abstain discard the temporary state. A unit test should hash the active parameters before and after each non-adapt decision and require equality.

H.2 Evidence Determinism

Evidence extraction must define preprocessing, batch order, numerical precision, class ordering, normalization constants, and missing-value behavior. Paired features require frozen and candidate outputs on the same samples in the same order. Source statistics and ATC thresholds are immutable calibration artifacts. Evidence schema versions should be explicit rather than inferred from vector length, because two schemas can share a dimension while changing semantics.

H.3 Calibration Identity

The calibration identity should hash the source checkpoint, candidate adapter class, adapted parameter names, optimizer and learning rate, update steps, evidence schema, benefit-model hyperparameters, feature scaler, residual construction, α, and inference-unit definition. The radius belongs to that identity. Reusing it after a candidate or evidence change is not a benign software optimization; it changes the random variable whose residual is being covered.

H.4 Runtime Accounting

A deployment report should time capture/preprocessing, frozen inference, candidate adaptation, candidate inference, evidence extraction, benefit prediction, interval decision, copy creation, and rollback separately. Mean and tail latency should be reported at the decision-window level. Peak memory should include both

Table 18: Feature families used by the maintained implementation. All quantities are aggregate statistics of unlabeled inputs, frozen/candidate outputs, or the proposed update.

Family	Representative coordinates	Required state	Failure handling
Predictive uncertainty	mean entropy, confidence, quantiles	frozen/candidate softmax	reject NaN or malformed probabilities
Marginal balance	normalized class-marginal entropy and change	predicted-class marginals	fixed class count and normalization
Model change	entropy gap, confidence drop, prediction disagreement	paired frozen/candidate outputs	require aligned sample order
Collapse signal	fraction above adapted confidence 0.9	candidate probabilities	threshold fixed before scoring
Distribution shift	free-energy shift, batch/running-statistic divergence	frozen logits and source statistics	missing source statistic forces non-commitment
Update magnitude	parameter-update ℓ_2 norm	shadow candidate delta	configuration hash must match calibration
Source-calibrated proxy	ATC-style threshold estimate	source-fixed threshold	target labels never used

model states and optimizer buffers. The compact manuscript leaves these cells unfilled because the canonical cross-dataset manifest does not provide one comparable runtime measurement. Omitting an unsupported number is preferable to quoting controller-only latency as end-to-end cost.

I Evidence Inventory

I.1 Protocol-Specific Feature and Model Schemas

The controlled-grid implementation records 11 label-free coordinates with effective rank 9. Two coordinates are deterministic differences: `entropy_drop=pre_entropy-post_entropy` and `pbal_drop=pre_pbal-post_pbal`. The panel therefore contains 11 stored values, not 11 independent information sources. Together they describe confidence, class balance, change from the frozen model, and update size.

Natural-shift artifacts do not all use the same schema. iWildCam, RxRx1, and most archived tracks use the 11-feature panel. Camelyon17 uses 17 features by adding disagreement, entropy gap, free-energy shift, batch-statistic divergence, a source-fixed ATC estimate, and confidence drop. Office-Home uses 18 features and also records source-prior KL and mean candidate disagreement. A model and its radius are valid only for the schema on which they were calibrated. Missing, reordered, or invalid features force non-commitment.

The controlled and archived GBRT protocols use scikit-learn's `GradientBoostingRegressor` Pedregosa et al. (2011) with squared-error loss, 250 trees, depth 2, learning rate 0.05, subsample 0.8, and random seed 0. The fit is specific to each dataset and adapter. CCT-20 instead uses its sealed ridge model, as described in Section 7.2. Thus "adapter agnostic" describes the interface: any adapter can propose a candidate that is evaluated and either committed or rolled back. It does not mean that a benefit model fitted for Tent transfers validly to EATA, SAR, or another adapter.

The base panel has 11 coordinates; the canonical Camelyon17 rich-Z artifact has 17 and the Office-Home artifact has 18. Other natural tracks retain the 11-coordinate base panel. These protocol-specific dimensions are part of the fitted estimator contract. They should not be described as universal sufficient statistics: a different shift may hide the benefit direction from all listed features. Feature selection can improve empirical prediction, but cannot defeat the label-kernel construction of Lemma 1 over an unrestricted target fibre.

J Failure-Safe Deployment Checklist

The audit log should store a timestamp, dataset or stream identifier, checkpoint and adapter hashes, evidence schema version, evidence vector, benefit estimate, radius, interval endpoints, action, rollback outcome, latency components, and configuration hash. Labels, if later revealed, are joined in a separate offline audit table. This separation makes it possible to verify that the live decision was target-label-free.

Table 19: Operational failures and conservative responses.

Failure	Detection	Response
Checkpoint/adapter mismatch	stored hash mismatch	rollback and recalibrate
Evidence outside support	distance or support audit	abstain
Missing/invalid feature	schema and finite-value check	retain f_0; abstain
Insufficient calibration size	rank $k > n$	infinite radius
Batch too small	minimum-size contract	wait or abstain
Residual transfer concern	group/residual diagnostic	widen or withhold claim
Continual-state drift	state and stream monitor	restore last certified state

K Prospective Natural-Shift Confirmation Protocol

The sealed CCT-20 study supplies a completed natural-shift target evaluation. Its 45 checkpoint–location cells contain 1 helpful and 44 harmful candidate updates, but KGA makes 44 FREEZE, 1 ABSTAIN, and 0 ADAPT decisions. It therefore matches always-freeze and has lower regret than always-adapt, providing evidence of conservative harm avoidance under natural shift. The predeclared stronger routing criterion was not met because the policy never selected ADAPT and did not select the single helpful cell. A second locked attempt on So2Sat-LCZ42 stopped at its development-feasibility gate with no eligible Tent or SAR candidate. No held-out gate-calibration rows or target inputs were accessed. The following protocol therefore describes a new versioned confirmation, not a continuation or reinterpretation of v1.

All So2Sat v1 gate-fit outcomes are open and cannot be used to tune a fresh confirmatory arm. Its target protocol also defines one action per city, whereas the current runner creates one action per city–checkpoint cell. That mismatch did not affect the development stop, but the target runner must remain disabled until a versioned lock resolves the action unit. The design below is future evidence and is not counted as a completed target experiment in this paper.

K.1 Lock Before Target Labels

The protocol should name one source checkpoint, one candidate adapter, one evidence schema, one benefit-model class, one calibration rule, one α, and one target-domain family. Adapter hyperparameters must be selected on development domains and then frozen. The lock should hash the checkpoint, code revision, configuration, calibration records, and planned scoring script. It must also name the test unit and the multiplicity family. A later change creates a new protocol version; it does not overwrite an earlier development-stopped or incomplete arm.

Natural heterogeneity should be built through independently defined environments rather than by pooling whichever benchmark conditions produce opposite signs. Examples include hospitals, camera locations, laboratory batches, acquisition days, physical devices, or visual domains. Environment membership may be observed at deployment, but the measured cell-benefit label remains hidden. The target test environments should be absent from estimator fitting, residual calibration, threshold tuning, and candidate selection.

K.2 Three-Way Data Separation

Development data fit h_θ and select the evidence pipeline. A separate calibration set produces residuals for Eq. (10). The held-out target set is opened exactly once after all software and configuration hashes match the lock. Splitting adjacent frames, patches, or samples from the same session across these stages is leakage even when filenames differ. The split must occur at the environment or session level that defines deployment exchangeability.

The target stream is replayed identically for always-freeze, always-adapt, simple gates, KGA without a radius, and the full certificate. Model randomness should either be frozen for paired replay or repeated through independently trained checkpoints declared in advance. Every policy receives the same ordered windows. Labels are stored behind a separate audit interface and are unavailable to the live decision process.

Table 20: Minimum lock for a confirmatory natural-shift KGA experiment.

Component	Must be fixed	Audit evidence
Model and candidate	checkpoint, adapter, optimizer, steps, adapted parameters	hashes and immutable configuration
Evidence and estimator	schema, normalization, model class, hyperparameters, seed	feature contract and fitted-artifact hash
Calibration	split unit, residual construction, exact rank, α	calibration IDs, residuals, radius, no target overlap
Target evaluation	environments, windowing, order, exclusion rules, transductive/inductive candidate semantics	sealed manifest and one-time scoring log
Primary inference	two paired regret gaps, cluster unit, interval method	code and random seed fixed before label reveal
Safety	$\mathrm{FA_u}$ and adapt exposure	action log joined to labels only offline
Multiplicity	all candidate–track arms in family	time-stamped inventory including failures
Runtime	end-to-end components and peak memory	raw timing rows on the target hardware

K.3 Endpoints and Decision Rule

The primary endpoint is paired regret against each fixed policy. Success requires both baseline minus KGA regret gaps to be positive and both predeclared intervals to exclude zero, with coverage justified for the declared estimand and sampling model. If the intended claim concerns population risk rather than measured cells, the protocol must also justify that transfer. The false-adapt endpoint is $\mathrm{FA_u}$ with the adapt count reported beside it. Secondary endpoints are $\mathrm{FA_c}$, adapt rate, freeze rate, abstain rate, commitment rate, task accuracy, and latency. Conditional false-adapt is descriptive unless a separate selective-risk result is proved.

The primary certificate rule remains unchanged: adapt if $\widehat{\Delta} - \varepsilon > 0$, freeze if $\widehat{\Delta} + \varepsilon < 0$, and abstain otherwise. The experiment must not lower ε, change α, or select a different benefit regressor after observing target labels. Exploratory sensitivity analyses may be reported, but they remain outside the primary bar.

K.4 Interpreting Outcomes

Four outcomes are scientifically useful. An interval-supported reduction in regret relative to each fixed comparator, with nontrivial ADAPT and FREEZE exposure, supports routing utility under the locked natural-shift class. Matching the better fixed policy with few errors supports conservative commitment but not selective routing. High abstention with low error indicates a conservative certificate and motivates better evidence or more calibration data. A comparator shortfall fails the predeclared routing-utility criterion; excess false adaptation challenges the corresponding safety or calibration criterion. Both outcomes must remain in the result family. None of these outcomes alone establishes the population band: that conclusion still requires the declared class and its structural assumptions.

L Reviewer Guide to Theorem Dependencies and Nonclaims

The compact paper uses three related logical statements. First, the disagreement reduction is an identity under the binary setup. Second, the population frontier is a uniform statement over a rich, declared class. Third, the empirical certificate is a marginal implication from interval coverage. The population statements concern Δ; the reported empirical calibration concerns Δ^{cell}. Connecting these targets requires an additional validity argument. Neither the assumptions nor the quantifiers are interchangeable.

Why the boundary is separate. For $|M| < \beta$, the construction chooses two latent residuals that yield opposite nonzero benefits. At $|M| = \beta > 0$, one admissible residual can make the benefit zero while another retains the sign of M; opposite nonzero signs need not exist. Both cases block a uniform strict commitment, but for different reasons. At $M = \beta = 0$, every law in the zero-residual class has zero benefit under the model. Abstention remains the maximal action only under the paper's strict-direction convention, not because either task-risk choice is worse.

Table 21: Dependency map. A check in one row does not validate premises in another row.

Statement	Inputs	Conclusion	Not concluded
Disagreement reduction	binary labels, 0/1 loss, $\mu_T(D) > 0$	$\Delta = 2\mu_T(D)(M + \gamma)$	calibration transfer or observability of γ
Interior impossibility	label-free fibre, $\beta > 0$, $\|M\| < \beta$, class richness	evidence-identical laws with opposite nonzero benefits	every empirical abstention is structural
Closed-band result	strict directional semantics and boundary law	no uniformly sound strict action on $\|M\| \leq \beta$	adapt/freeze have different task risk at $\Delta = 0$
Frontier sufficiency	declared $\|\gamma\| \leq \beta$ and fixed M	direction is fixed when $\|M\| > \beta$	label-free estimation of β
Coverage-to-action proposition	marginal interval coverage for a declared target B	marginal false-adapt/freeze control for B	conditional error control or risk alignment
KGA benchmark	fitted h_θ, declared residual rule, frozen protocol	measured regret and actions on saved units	population-benefit coverage or universal transfer to a new domain
Lean build	encoded definitions and assumptions	kernel-checked formal statements	correctness of unencoded data and deployment assumptions

Why a calibration residual need not be drift. Let $P_T = P_S$, with equiprobable inputs a, b, deterministic labels $Y(a) = 1$, $Y(b) = 0$, and candidate $f_a \equiv 1$. Set $f_0(a) = 0$, $f_0(b) = 1$, and $s \equiv \frac{1}{2}$. The candidate is correct on half the source inputs, so this constant score is source-calibrated in the ordinary sense. Yet disagreement occurs only at a, where the candidate is always correct. Thus $M = 0$, $\gamma = \frac{1}{2}$, and $\Delta = \frac{1}{2}$, although the source and target laws are identical. Conditioning on disagreement, not a distribution change, explains the nonzero residual. The frontier requires a bound on this residual regardless of its cause.

Why coverage is the premise. The certificate proof is elementary once Eq. (9) holds: on the coverage event, an interval wholly above zero cannot contain a nonpositive Δ^{cell}. The scientific burden is therefore to justify coverage for the declared outcome and calibration/deployment pair. Joint exchangeability of the residual scores is one route; known shift correction or another valid construction may be another. Risk alignment can explain why a useful predictor transfers, but it is not an additional logical premise after coverage is given.

Why cell coverage is not population coverage. Consider the fixed binary predictors $f_0 \equiv 0$ and $f_a \equiv 1$. Let the target input be a with probability 0.45 and b with probability 0.55, with deterministic labels $Y(a) = 1$ and $Y(b) = 0$. Population benefit is $\Delta = -0.10$. In a one-item evaluation cell, a fixed predictor that maps a to $+1$ and b to -1 predicts Δ^{cell} perfectly without reading the new label. With radius zero, cell coverage is perfect, yet the gate selects ADAPT with probability 0.45 despite negative benefit under the fixed population law. This constructed example does not contradict Proposition 2; it shows why its covered target cannot be changed.

Why exchangeability alone does not validate sign flips. As a separate null example, let all nine cluster gaps equal the same fair random sign S. This vector is exchangeable, each gap is symmetric about zero, and a global sign change preserves its law. Nevertheless, when $S = +1$, independent-coordinate sign-flip enumeration for the mean gives a reference value of $1/512$. That occurs with probability $1/2$, not $1/512$. The vector fails Eq. (12). This example explains the required assumption; it is not a model fitted to the CCT-20 observations.

Why a null is not an impossibility result. A null benchmark row can arise because the true effects are small, the candidate is one-sided, the evidence map is weak, the benefit model underfits, the radius is wide, or calibration does not transfer. The population impossibility requires evidence-identical target laws inside the declared class with incompatible strict directions. The canonical benchmark table does not estimate that full fibre. It therefore reports weak or adverse empirical regimes without calling them proven unknowable worlds.

What would falsify the practical claim. The practical claim is conditional and testable. A clean target replay with the locked pipeline can show excessive FA_{u}, poor coverage, or regret above both fixed policies. A checkpoint or adapter change can show that calibration does not transfer. A family-held-out split can show that cell-level cross-fitting was optimistic. Such failures do not refute the population identity; they refute the operational assumptions used to obtain a useful empirical interval.

M Formalization and Artifact Scope

The Lean project under `docs/research/kbound/formal/` provides a kernel-checked inventory for the theorem names it exports. The compact paper relies on the LaTeX statements above; it does not claim that every sentence has a one-to-one Lean theorem.

In particular, deployment-class richness, risk alignment of a learned evidence map, exchangeability of a benchmark split, and correctness of data preprocessing remain outside the proof assistant unless separately encoded.

The registered audit contains 142 declarations: the 65-result finite and algebraic core plus 77 probability capstones and counterexample results. The added modules derive one-shot residual coverage from exchangeable measurable scores, including ties; filtered Ville and bounded optional-stopping bounds; general KL/TV testing bounds for finite product experiments; and independent bounded and martingale-difference concentration inequalities. Their assumptions are explicit. In particular, these results do not establish exchangeability or conditional nulls for the benchmark data, nor do they turn batch-level coverage into population-risk coverage.

The measurable target-law construction also goes beyond the original two-atom example. It fixes the input law and off-disagreement label kernel, varies a measurable correctness field on disagreement, and integrates the actual zero-one loss difference. With labels supported on the two predictions on disagreement, feasible margins, and positive disagreement mass, it proves the clipped identified interval and exact strict ADAPT/FREEZE frontiers without assuming `RichAt`. It does not prove richness for an arbitrary restricted deployment subclass.

One historical extension remains outside the verified scope. A measurable label swap preserves label-free evidence and reverses population benefit, but selecting one representative from each swap orbit need not identify the sign: distinct orbits can share the same evidence law and have opposite selected signs. Lean verifies this counterexample and the corresponding set-theoretic fibre-consistency criterion, not a general measurable decoder or the older H/ratio-rate extension. Those stronger claims are excluded from this compact submission. The `-full-foundations` gate therefore remains failing for the sixth layer; the scoped kernel audit checks the registered results and rejects nonstandard proof axioms.

The formal build and empirical build answer different questions. Lean checks that a proposition follows from formal assumptions. The reconciliation scripts check that displayed aggregates match saved records under implemented definitions. Neither checks the truth of an unencoded deployment assumption, and neither substitutes for independent experimental replication.

The authoritative compact artifacts are therefore layered:

1. this source and its compiled PDF for the scoped narrative;
2. shared theorem inputs for the exact statements and proofs;
3. canonical JSON and generated LaTeX for numerical results;
4. the source manifest for record-level provenance;
5. package tests and Lean logs for implementation and formal checks.

Historical drafts, dashboards, and exploratory search folders can remain useful development records, but they are not evidence sources for this compact submission unless referenced by the canonical manifest.

N Reproducibility and Release Procedures

N.1 Canonical Numerical Source of Truth

The CIFAR primary rows and the supplementary reconciled point estimates have one numerical source: `experiments/kbound/results/reconciled_panels_v1/canonical_panel_results.json`. Its compact source manifest has SHA-256 `266ce7cd87ae795e0742d43c40afefef82e82b555baa7dda8c6d270f73e2c0`

31. Named auxiliary artifacts support only their stated roles. The code-bound current-policy family sensitivity is retrospective and fails the confirmatory bar: Holm over six prospectively named contrasts gives adjusted retrospective sign-flip reference values of 0.09375 for both Tent comparisons, but the replay, corruption-family unit, and test were not preregistered. The earlier Tent cluster artifact and archived POEM/AETTA comparison remain historical. The So2Sat-LCZ42 selection artifact is a separate locked development-gate record with no promoted target row. None of these overwrites the canonical point-estimate panel. The generated result manifest indexes every promoted claim and its scope.

N.2 Reproduction Pipeline

The full gate requires Python 3.12 with the repository-wide research profile `requirements-research.txt`; the smaller paper-only profile does not include the Torch imports used during repository-wide test collection. The model runtime uses PyTorch Paszke et al. (2019). From the repository root, the release command is:

```
KBOUND_PYTHON=.venv/bin/python bash docs/research/kbound/runbooks/release_candidate.
sh all
```

It runs the read-only preflight, validates and regenerates result authorities, executes software and formal checks, builds the two PDFs and compact DOCX, renders every PDF page, and emits release checksums. It consumes frozen model outputs and never launches training.

For presentation-only rebuilding from already validated authorities, the command is:

```
BUILD_LONG_TMLR=1 BUILD_DOCX=1 bash docs/research/kbound/scripts/build_pdfs.sh
```

This shorter command validates the frozen authorities, regenerates presentation tables and figures, and builds the maintained PDFs and DOCX. It does not replace the full release gate or create new empirical evidence.

N.3 Calibration Identity and Implementation Contract

The maintained API binds a certificate to the checkpoint, adapter configuration, evidence schema, benefit estimator, residual construction, safety level, and scoring unit. The deployment path must fail closed when any identity differs. `KGA.certify_evidence` requires a frozen estimator, the exact schema, a protocol hash, and a separate residual set. The lower-level `certify` function is an audit interface; it does not construct evidence or validate transfer by itself.

N.4 Formalization and Lean Scope

Lean artifacts and validators check the encoded theorem cores and selected finite constructions. Formalization of learning-theory results has precedent, including a Rademacher-complexity generalization bound Sonoda et al. (2025); we cite its pinned 2025 version as context, not as a proof of K-Bound's assumptions. These checks do not certify dataset correctness, preprocessing, empirical transfer, exchangeability, or every deployment assumption stated in prose. Machine checking strengthens the statements it encodes; it does not validate the benchmark pipeline outside those statements.

N.5 Release Checklist and Audit Logs

The Phase-1 provenance audit matches all 106 canonical source/compact hash pairs, all 10 recoverable configuration hashes, and all 6 archived checkpoint hashes; it also records a 21-file current-code snapshot. Historical runs that did not serialize full identity remain explicitly partial rather than being backfilled. The release gate runs reconciliation tests, claim-ledger and result-manifest validation, manuscript checks, LaTeX builds, checksum verification, and visual PDF inspection. Build logs, the canonical manifest, the provenance seal, and the final PDFs are archived together.

Repository URLs and author-identifying metadata must be withheld or redacted for a double-blind venue. The local paths above describe artifact roles without providing a public identity link.